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Market Tremors: Analyzing the Impact of Geopolitical Events on Equity Indices

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Abstract

This project investigates the impact of the Russia–Ukraine conflict, which began with the invasion on February 24, 2022, on global equity markets. By analyzing the market behavior of key indices—including Russia's MOEX, Germany's DAX, and the U.S. S&P 500—along with commodity prices such as Brent Crude and the USD/RUB exchange rate, we aim to assess the immediate and short-term financial consequences of geopolitical escalation. Utilizing Bloomberg Terminal tools such as EVTS, GIP, GP, and HIST, we extract and examine key data points surrounding the conflict. This study applies a combination of descriptive statistics, return analysis, and volatility modeling to uncover patterns in market response, providing insights into how global markets react to geopolitical instability.

Keywords

Russia-Ukraine Conflict, Equity Markets, Bloomberg Terminal, Market Volatility, MOEX, DAX, S&P 500, Brent Crude, USD/RUB, Geopolitical Risk, Event Study, Financial Analysis

1. Introduction

The Russia–Ukraine conflict, which escalated in February 2022, triggered significant economic uncertainty and geopolitical volatility across global markets. As sanctions, supply chain disruptions, and energy security concerns intensified, the ripple effects were evident not only in Russian assets but across sectors such as aerospace, defense, energy, agriculture, and currency markets. This project aims to explore the short-term market response to this geopolitical shock, using a combination of return, volatility, and temporal pattern analysis across global indices and sector ETFs.

The study focuses on identifying how different asset classes and regions dynamically responded to the conflict, uncovering both immediate and lagged effects using tools such as volatility tracking and Dynamic Time Warping (DTW). The goal is to derive actionable insights on market structure shifts, lead-lag relationships, and sectoral decoupling that occurred during the initial months of the invasion.

2. Russia-Ukraine Conflict Background

On February 24, 2022, Russia launched a full-scale invasion of Ukraine, leading to one of the most severe military and humanitarian crises in Europe since World War II. The event prompted swift economic sanctions from the West, disrupted global supply chains, and caused immediate volatility in energy and financial markets. Investors reacted to the uncertainty by fleeing to safer assets, and several markets, particularly in Russia and Europe, saw sharp drawdowns. This conflict remains ongoing, continuing to influence commodity markets, defense spending, and geopolitical strategy.

3. Data Sources & Dataset Description / Methodology (Tools + Bloomberg Functions)

We used Bloomberg Terminal to extract historical market data using functions such as HIST, GP, and GIP. The dataset comprises daily historical data from January 25, 2022 to March 25, 2022 for a curated set of indices, sector-specific ETFs, individual defense and energy stocks, and currency pairs. Openly available data was also sourced from investing.com and tickertape.in for less accessible tickers (especially European and Indian defense firms).

Asset Coverage

Main Indices	MOEX, DAX, S&P 500 (SPY)
Defense and Aerospace	ITA, PPA, JETS
European/Indian Defense	RHMG, BAES, HAL.NS, PRAF.NS, BARA.NS, AIR, MTXGN
Commodities & Currencies:	Brent Crude (BRNT), USD/RUB, USD/UAH
Energy Sector	XLE, BP, ENI, SHELL, EQNR
European Equities	FEZ, IEUR, SXEPEX

Table 1: Key Indices and Assets to Study

Each asset has the following fields: Date: Trading day Price: Adjusted close price Change: Daily return (%) Volatility: 10-day rolling volatility computed over returns

The assets studied span a diverse set of global financial instruments, including equity indices (MOEX Index – Russia, DAX – Germany, S&P 500 – U.S.), commodities and currencies (Brent Crude Oil, USD/RUB, USD/UAH), aerospace and defense ETFs (ITA, PPA, JETS), individual European and Indian defense stocks (e.g., RHMG, MTXGN, BAES, HAL), energy sector stocks (e.g., BP, ENI, EQNR, SHELL), and broader regional equity ETFs (IEUR, SXEPEX, FEZ). Each dataset contains daily adjusted closing prices, calculated returns, and rolling volatility, spanning a 60-day window from January 25 to March 25, 2022 — covering 30 days before and after the Russian invasion of Ukraine on February 24, 2022. The data has been harmonized into a consolidated wide-format structure for cross-asset comparison.

Data to Extract

Component	Details
Event Date	February 24, 2024
Analysis Window	January 25 – March 25, 2022 (t–30 to t+30)
Assets	(refer table 1)
Metrics	Date (Trading Day), Adjusted Close Price, Volume, Daily Return (%), 10D Volatility
Bloomberg Functions	HIST, GP, CN, GIP

Table 2: Data to Extract

Derived metrics such as daily returns and 10-day rolling volatility were calculated to evaluate price fluctuations and market risk. A binary event marker was also added to separate pre- and post-invasion observations. Optional qualitative insights were gathered using Bloomberg news headlines (CN, TOP) and country-specific geopolitical risk scores (GIP).

4. Data Preprocessing Steps

4.1. Data Acquisition and Import

Data for each selected asset (equities, indices, commodities, currencies, and ETFs) was initially sourced from Bloomberg Terminal and exported in Excel format. Each file contained multiple columns including Open, High, Low, Close, and Volume, typically in varying formats depending on the asset and source.

```
df_moex.head()
```

	Date	Price	Open	High	Low	Vol.	Change %
0	2022-03-25	2,484.13	2,620.62	2,664.74	2,470.69	NaN	-3.66%
1	2022-03-24	2,578.51	2,467.09	2,761.17	2,447.68	NaN	4.37%
2	2022-02-25	2,470.48	2,261.33	2,552.46	2,256.09	NaN	20.04%
3	2022-02-24	2,058.12	2,735.88	2,740.31	1,681.55	NaN	-33.28%
4	2022-02-22	3,084.74	2,909.40	3,113.40	2,756.46	NaN	1.58%

Figure 4.1 (a): Example Raw Data Head (Moex)

```
df_moex.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25 entries, 0 to 24
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Date      25 non-null     datetime64[ns]
 1   Price     25 non-null     object
 2   Open      25 non-null     object
 3   High      25 non-null     object
 4   Low       25 non-null     object
 5   Vol.      0 non-null      float64
 6   Change %  25 non-null     object
dtypes: datetime64[ns](1), float64(1), object(5)
memory usage: 1.5+ KB
```

Figure 4.1 (b): Example Raw Data Information (Moex)

4.2. Column Filtering

Only the relevant columns were retained for analysis. Specifically:

- Retained: Date, Close, and Volume
- Dropped: Open, High, Low, and other metadata columns

This helped reduce dimensionality and focus the analysis solely on end-of-day trading metrics.

```
df_moex.head()
```

	Date	Price	Vol.	Change %
0	2022-03-25	2,484.13	NaN	-3.66%
1	2022-03-24	2,578.51	NaN	4.37%
2	2022-02-25	2,470.48	NaN	20.04%
3	2022-02-24	2,058.12	NaN	-33.28%
4	2022-02-22	3,084.74	NaN	1.58%

Figure 4.2 (a): Example Column Filtering (Moex)

4.3. Standardization of Column Names and Formats

To ensure consistency across all assets:

- Column names were standardized to a uniform structure such as Price_<ticker> and Vol_<ticker> for closing prices and volumes respectively.
- The Date column was converted to a datetime object with a consistent format (YYYY-MM-DD), and datasets were sorted chronologically.
- Ticker symbols were lowercased and cleaned of any special characters to simplify downstream merging and referencing.

```
df_moex.head()
```

	Date	Price_moex	Vol_moex	Change_moex
0	2022-03-25	2,484.13	NaN	-3.66%
1	2022-03-24	2,578.51	NaN	4.37%
2	2022-02-25	2,470.48	NaN	20.04%
3	2022-02-24	2,058.12	NaN	-33.28%
4	2022-02-22	3,084.74	NaN	1.58%

Figure 4.3 (a): Example Column Name Standardization (Moex)

```
df_moex.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 25 entries, 24 to 0
Data columns (total 5 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Date              25 non-null     datetime64[ns]
 1   Price_moex        25 non-null     object
 2   Vol_moex          0 non-null      float64
 3   Change_moex       25 non-null     float64
 4   Volatility_moex   16 non-null     float64
dtypes: datetime64[ns](1), float64(3), object(1)
memory usage: 1.2+ KB
```

Figure 4.3 (b): Example Column Format Standardization (Moex)

4.4. Time Alignment on Trading Days

Since different markets operate on different holidays and weekends:

- All datasets were merged on the Date column using an inner join strategy to ensure consistency in temporal comparisons.
- Only dates that had valid trading data for all selected assets were retained in the unified master dataset.

4.5. Daily Return Calculation

For each asset:

- Daily returns were calculated as percentage change in adjusted closing price using the formula:

$$\text{Return}_t = \frac{P_t - P_{t-1}}{P_{t-1}} \times 100$$

- Columns were named as Change_<ticker> and stored in the master dataframe.

4.6. Rolling Volatility Computation

A 10-day rolling window was applied to the return series of each asset to estimate short-term volatility using:

- Standard deviation over the previous 10 trading days
- Resulting in a new column: Volatility_<ticker>

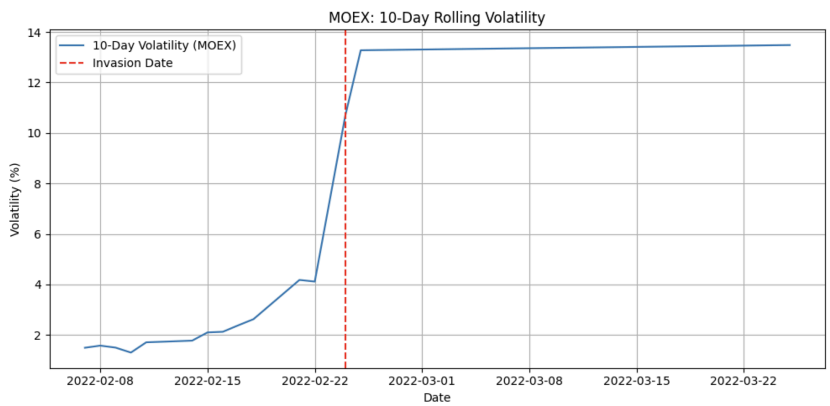


Figure 4.6 (a): MOEX 10-Day Rolling Volatility

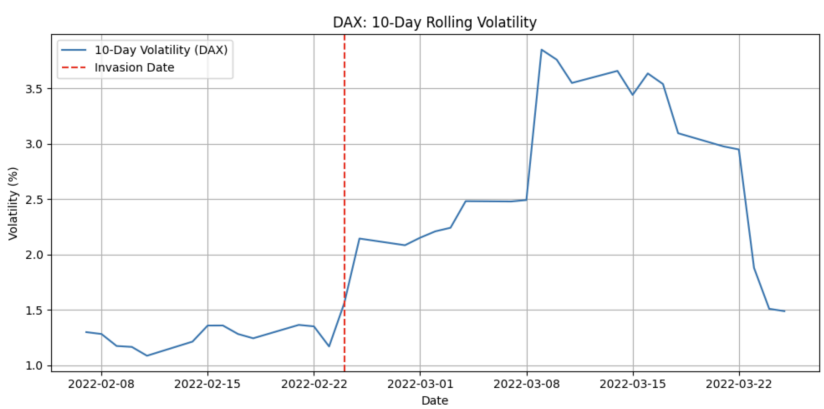


Figure 4.6 (b): DAX 10-Day Rolling Volatility

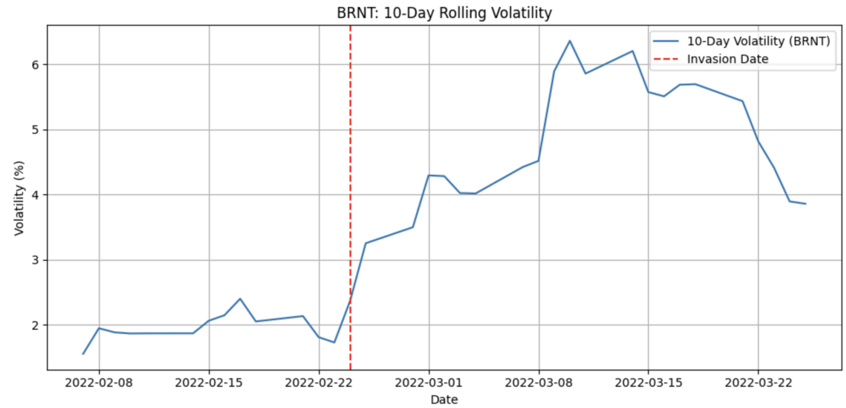


Figure 4.6 (c): BRNT 10-Day Rolling Volatility

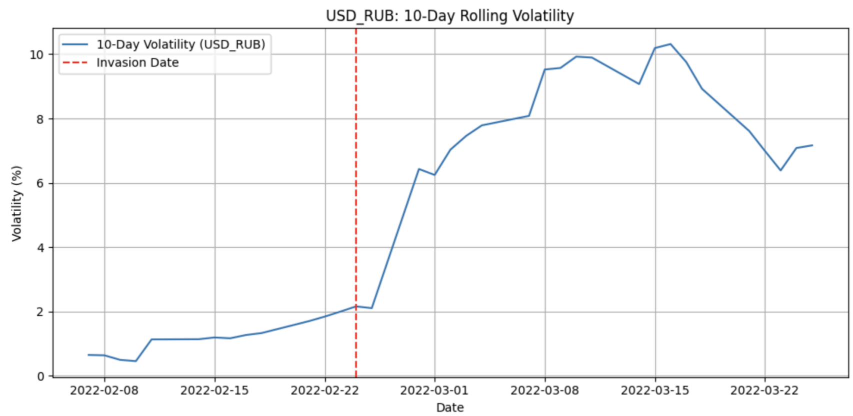


Figure 4.6 (d): USD_RUB 10-Day Rolling Volatility

This provided insights into market risk and panic levels during the study period.

4.7. Master DataFrame Construction

All asset-specific dataframes (after cleaning and feature engineering) were merged into a single wide-format master_df, containing:

- A unified Date column
- Daily Price_, Change_, Volatility_, and Vol_ columns for each asset

```python
# Display the first few rows of the master dataframe
display(master_df.head(2))
```

	Date	Price_moex	Vol_moex	Change_moex	Volatility_moex	Price_dax	Vol_dax	Change_dax	Volatility_dax	Price_spy	...	Volatility_eni	Price_eqnr	Vol_eqnr
0	2022-01-25	3,258.74	NaN	0.73	NaN	15,123.87	86.07M	0.75	NaN	434.47	...	NaN	249.8	4.51M
1	2022-01-26	3,357.66	NaN	3.04	NaN	15,459.39	82.97M	2.22	NaN	433.38	...	NaN	258.3	6.43M

2 rows × 102 columns

```python
master_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 44 entries, 0 to 43
Columns: 102 entries, Date to Post_Invasion
dtypes: datetime64[ns](1), float64(74), int64(1), object(26)
memory usage: 35.2+ KB
```

Figure 4.7 (a): Merging Datasets by Date to Create a Master Dataframe

This format allowed for easy comparison, correlation analysis, and matrix computations.

4.8. Post-Invasion Indicator

A binary flag column Post_Invasion was added to segment the dataset into two temporal zones:

- 0 → Pre-invasion (before February 24, 2022)
- 1 → Post-invasion (on or after February 24, 2022)

Vol_shell	Change_shell	Volatility_shell	Post_Invasion
1.88M	1.28	NaN	0
2.82M	5.38	NaN	0

Vol_shell	Change_shell	Volatility_shell	Post_Invasion
3.80M	3.74	2.276097	1
5.46M	-1.40	2.364958	1

Figure 4.7 (a): Merging Datasets by Date to Create a Master Dataframe

This enabled pre/post comparison in returns, volatility, and drawdowns.

4.9. Data Type Enforcement & Error Handling

To avoid runtime issues and ensure numeric computations:

- Key columns such as Price_moex and Price_dax were explicitly converted to float using pd.to_numeric(..., errors='coerce')
- Any parsing errors were handled by coercing invalid entries to NaN, which were later dropped or forward-filled as needed.

4.10. Volume Column Reduction

Many volume columns (e.g. Vol_moex, Vol_usd_rub) contained all-null or static values:

- These non-informative columns were dropped to streamline analysis and improve visualization clarity.

This robust preprocessing pipeline ensured that the dataset was clean, consistent, and well-structured for advanced analytics like Dynamic Time Warping, volatility clustering, and drawdown analysis.

5. Visual Analytics and Their Interpretation

To quantify and visualize the financial market response to the Russia–Ukraine conflict, we employed a series of structured data analytics techniques. These covered pre/post-event comparisons, volatility and drawdown analysis, temporal correlation dynamics, sector-level impact evaluation, and time-aligned behavior modeling via Dynamic Time Warping (DTW).

5.1. Pre- vs. Post-Invasion Return and Volatility Comparison

To assess the immediate market response to the conflict:

- A binary flag (Post_Invasion) segmented the dataset into pre-event (before Feb 24, 2022) and post-event periods.

- For each asset, the average daily return and average 10-day rolling volatility were calculated separately for both periods.
- The results were summarized in a table df_summary and visualized using bar charts, which made it easy to compare shifts in performance and risk across different assets.

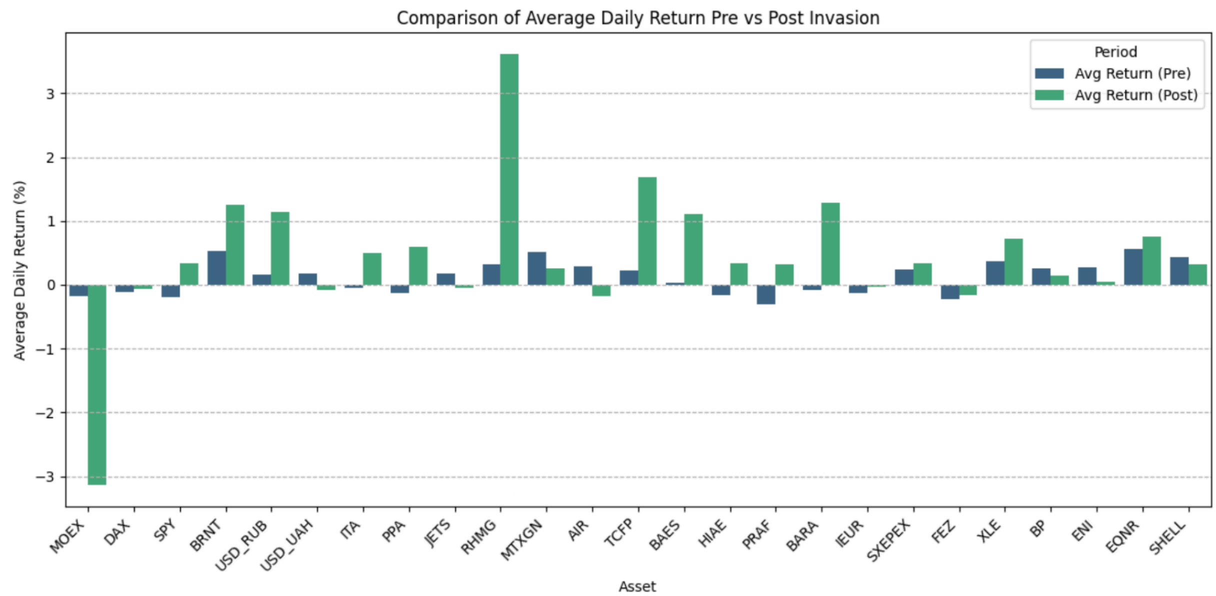


Figure 5.1 (a): Comparison of Average Daily Return Pre vs Post Invasion

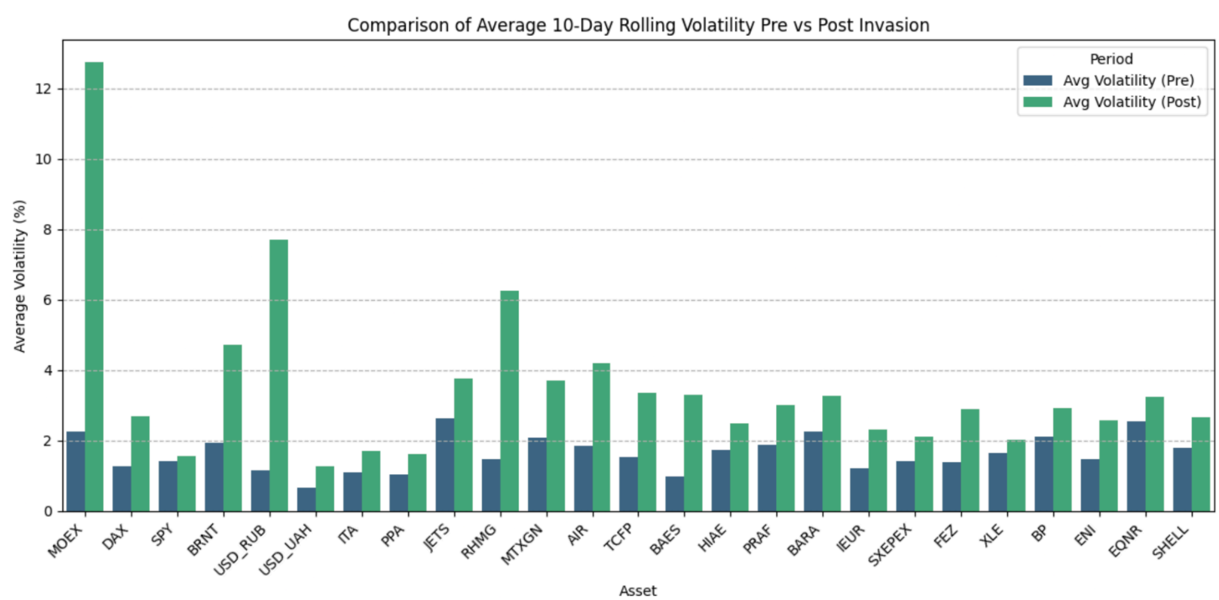


Figure 5.1 (b): Comparison of Average 10-Day Rolling Volatility Pre vs Post Invasion

This revealed that defense-related assets saw increases in average return post-invasion, while broad market indices and commodities experienced elevated volatility.

5.2. Volatility Spikes and Drawdown Analysis

To detect moments of market panic and asset fragility:

- The 10-day rolling volatility of each asset was plotted to observe short-term spikes around the invasion date.
- The maximum drawdown was computed for each asset using:

$$\text{Max Drawdown} = \min\left(\frac{P_t - \max(P_{1\ldots t})}{\max(P_{1\ldots t})}\right) \times 100$$

- This metric helped quantify the worst peak-to-trough decline experienced by each asset in the analysis window.

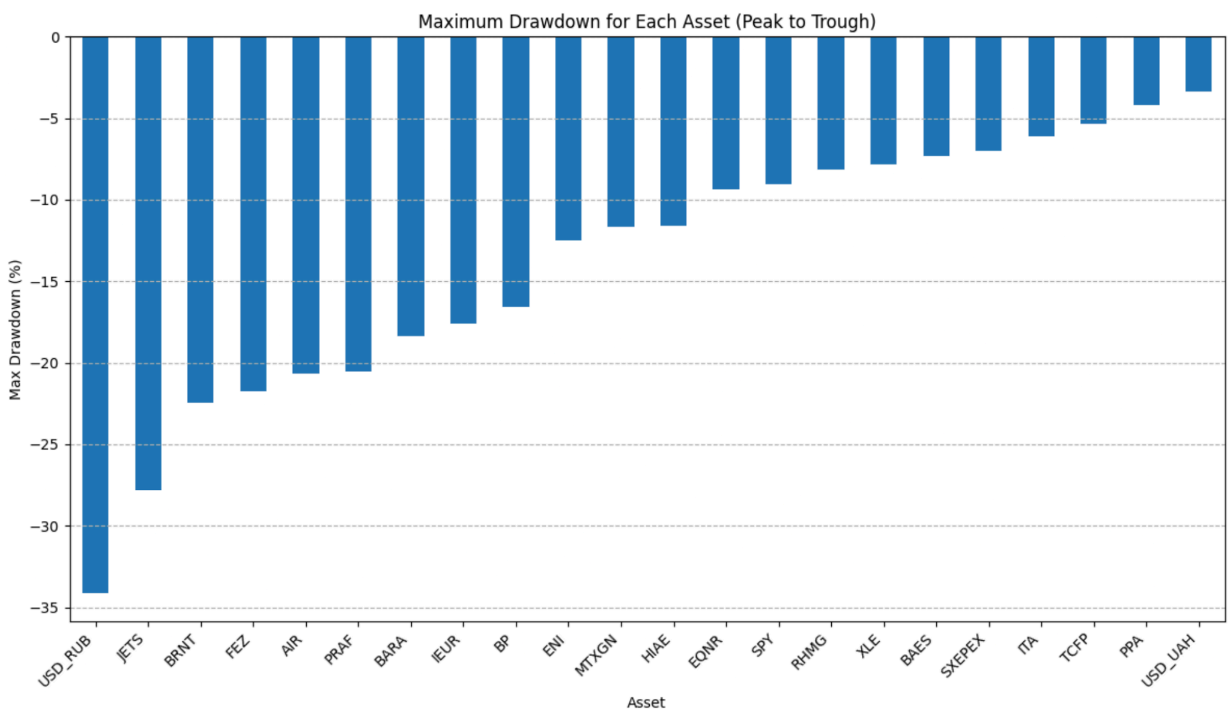


Figure 5.2 (a): Maximum Drawdown for Each Asset (Peak to Trough)

The results were displayed in a bar chart sorted by drawdown magnitude, revealing that MOEX (not shown due to insufficient data) and USD/RUB experienced the sharpest collapses, whereas defense stocks remained relatively insulated.

5.3. Correlation Matrix of Returns and Volatility

To understand interdependencies and contagion effects:

- Pairwise Pearson correlations were computed for:
    - Daily returns (Change_ columns)
    - 10-day rolling volatility (Volatility_ columns)
- Two separate heatmaps were generated to visualize these relationships.

This analysis uncovered strong correlations within sectoral and regional groups (e.g., SPY & DAX, ITA & PPA), and highlighted relatively weak or negative relationships between commodities and equities (e.g., Brent and MOEX).

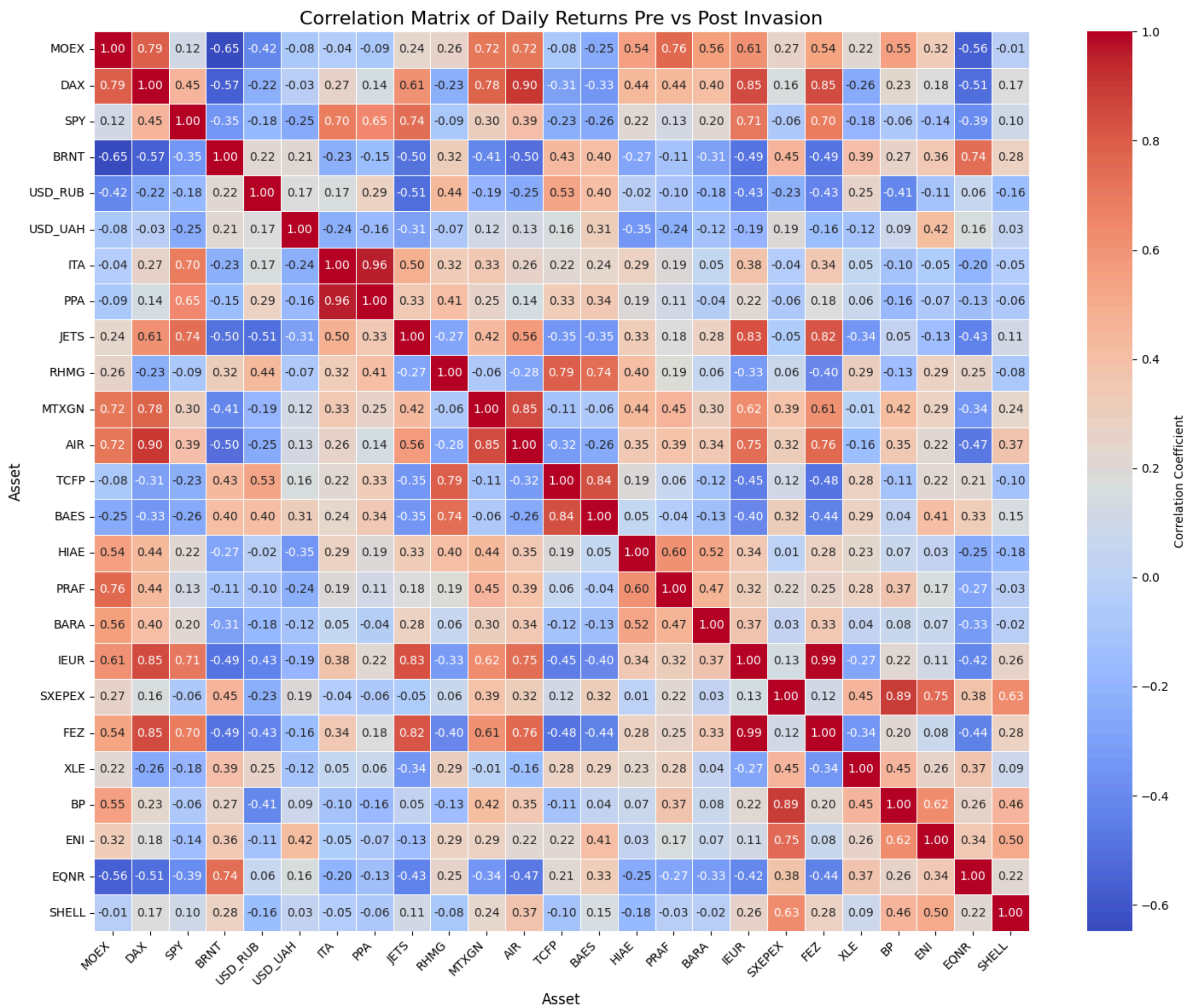


Figure 5.3 (a): Correlation Matrix of Daily Returns Pre vs Post Invasion

Additionally, separate correlation matrices were generated for pre- and post-invasion periods, showing how traditional correlations broke down or intensified during the crisis.

5.4. Rolling Correlation Analysis

To observe how relationships between key assets evolved over time:

- 10-day rolling correlations were computed for selected asset pairs (e.g., SPY vs. Brent, ITA vs. MOEX).
- The rolling correlation series were plotted to identify time-varying market co-movement patterns.

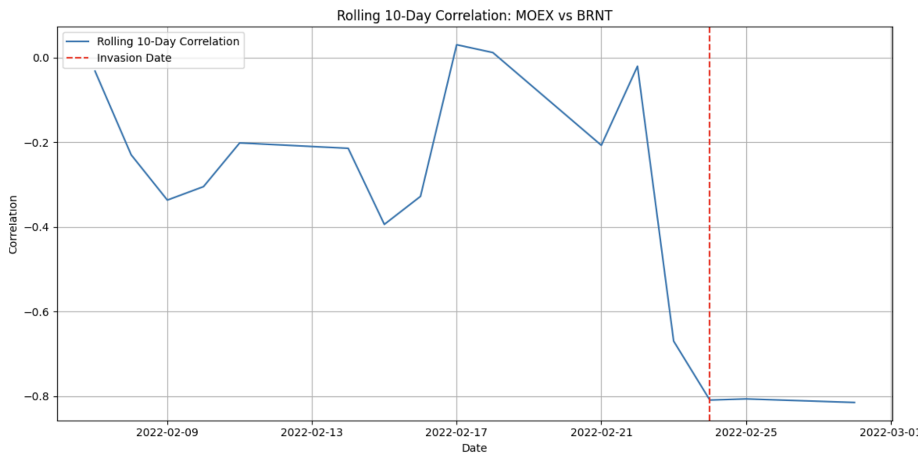


Figure 5.4 (a): Rolling 10-Day Correlation MOEX vs BRNT

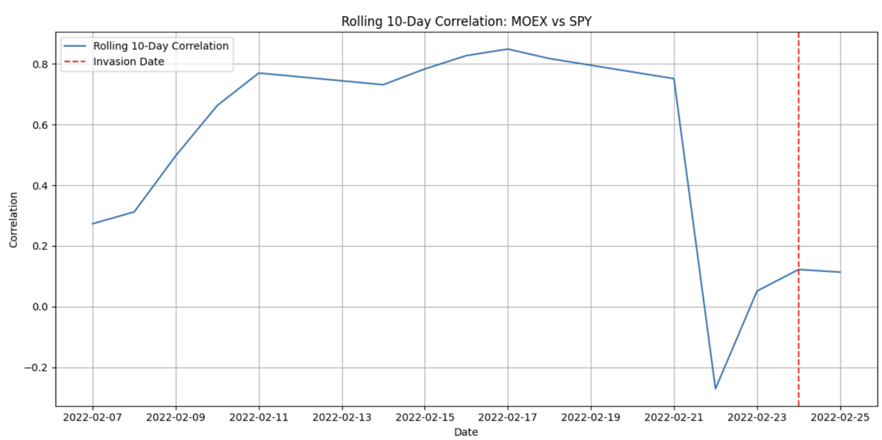


Figure 5.4 (b): Rolling 10-Day Correlation MOEX vs SPY

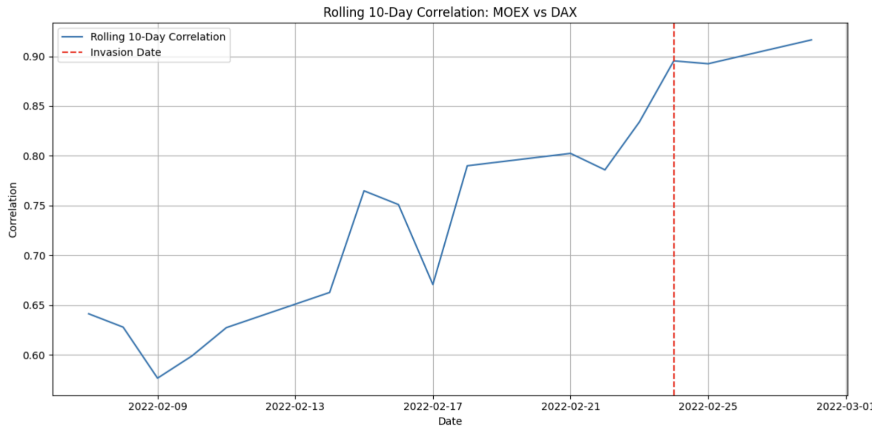


Figure 5.4 (c): Rolling 10-Day Correlation MOEX vs DAX

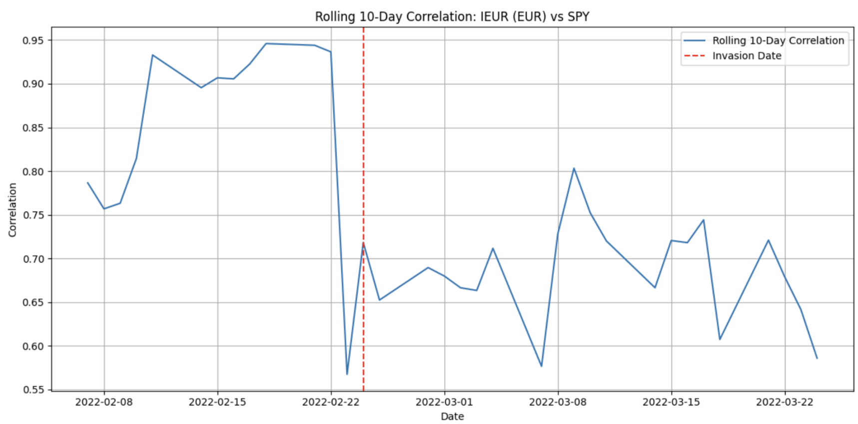


Figure 5.4 (d): Rolling 10-Day Correlation IEUR (EUR) vs DAX

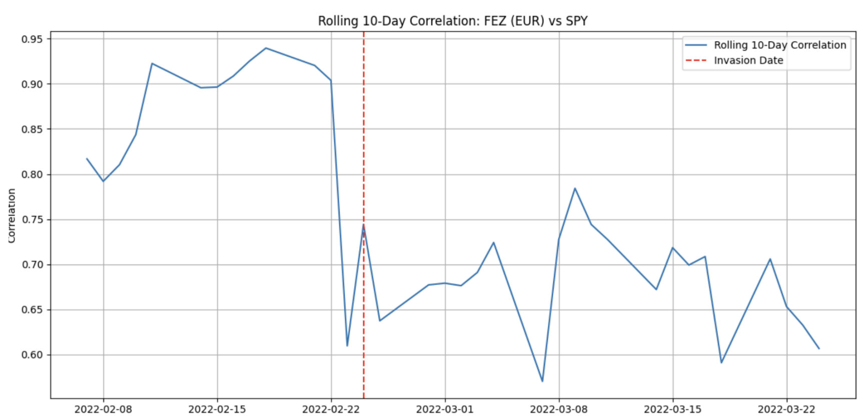


Figure 5.2 (e): Rolling 10-Day Correlation FEZ (EUR) vs SPY

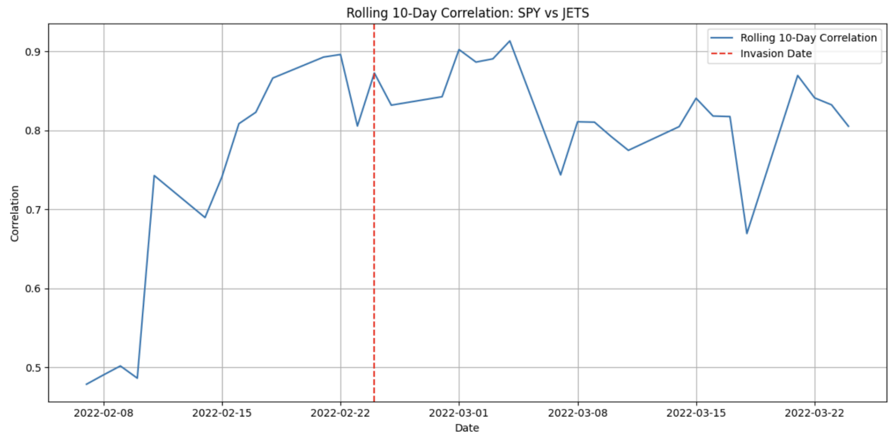


Figure 5.4 (f): Rolling 10-Day Correlation SPY vs JETS

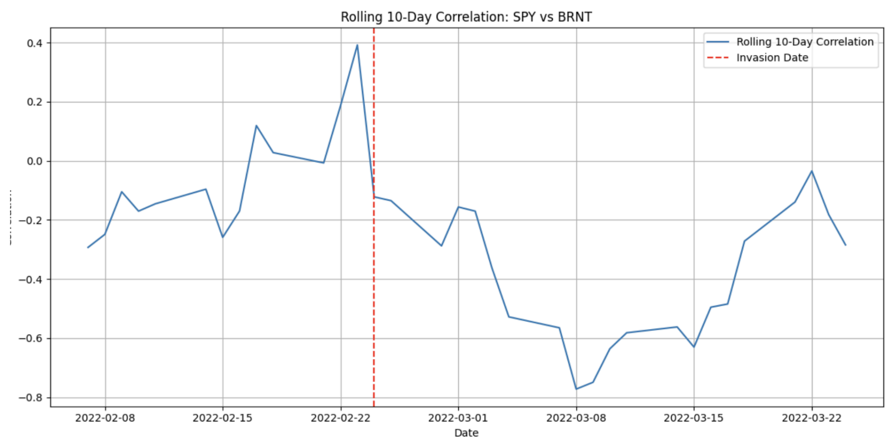


Figure 5.4 (g): Rolling 10-Day Correlation SPY vs BRNT

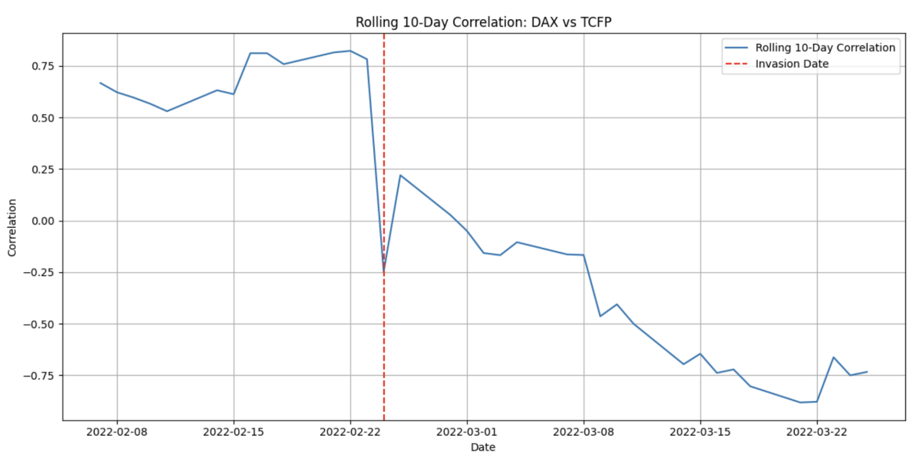


Figure 5.4 (h): Rolling 10-Day Correlation DAX vs TCFP

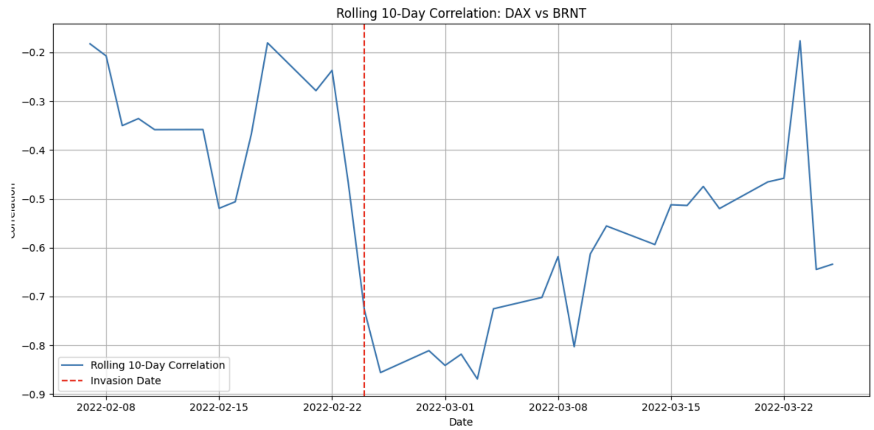


Figure 5.4 (i): Rolling 10-Day Correlation DAX vs BRNT

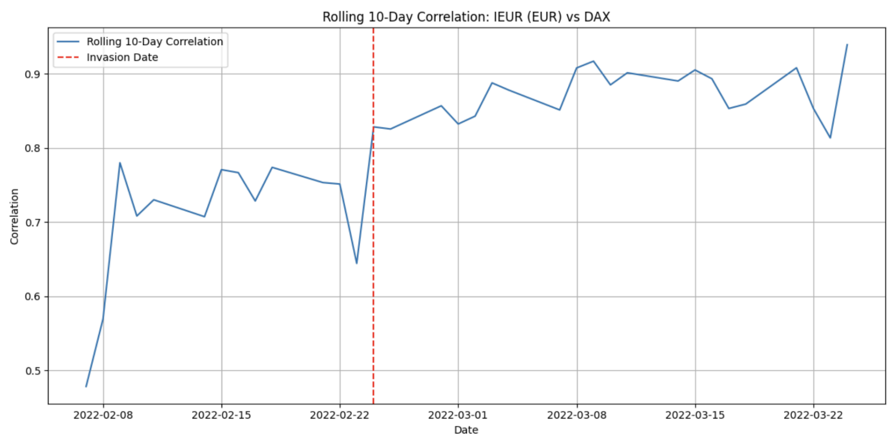


Figure 5.2 (j): Rolling 10-Day Correlation IEUR (EUR) vs DAX

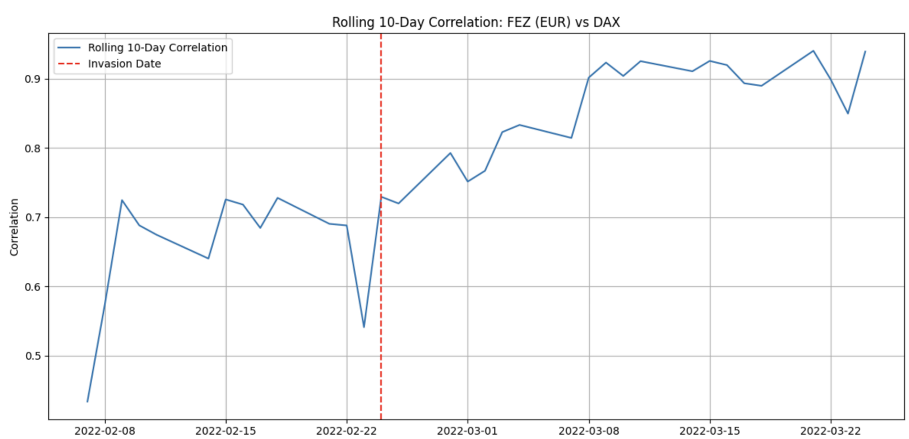


Figure 5.2 (k): Rolling 10-Day Correlation FEZ (EUR) vs DAX

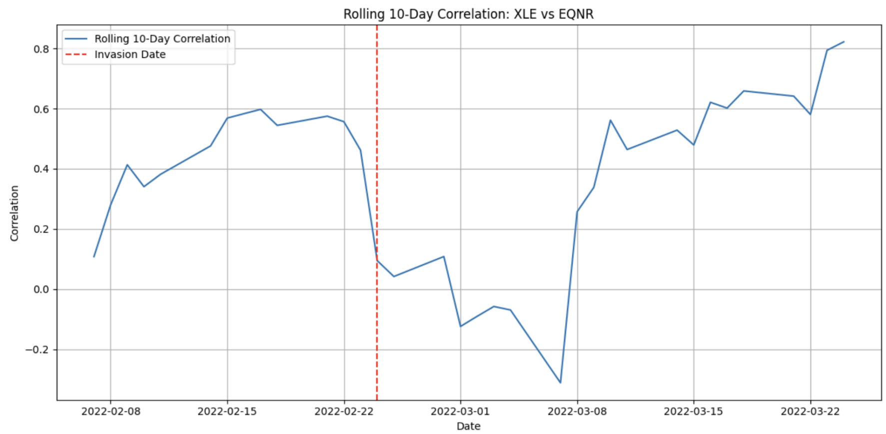


Figure 5.2 (l): Rolling 10-Day Correlation XLE vs EQNR

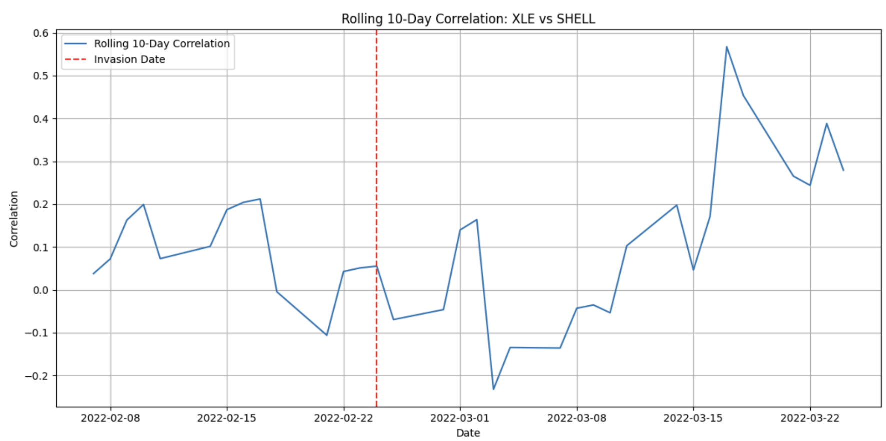


Figure 5.2 (m): Rolling 10-Day Correlation XLE vs SHELL

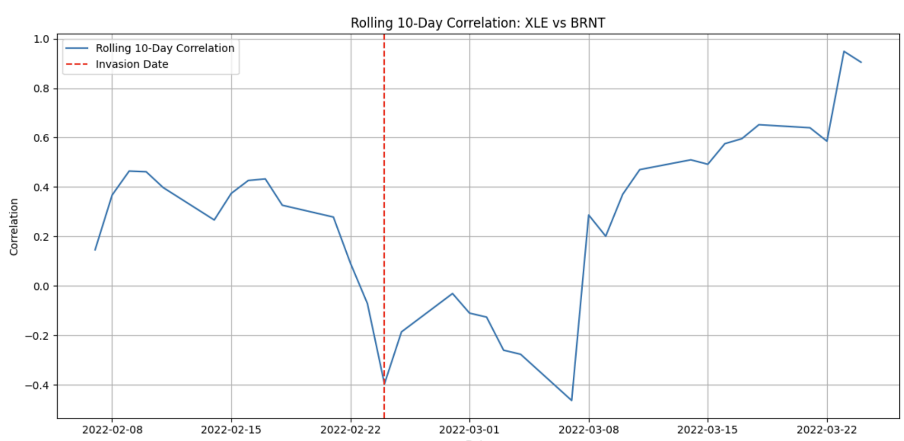


Figure 5.4 (n): Rolling 10-Day Correlation XLE vs BRNT

This dynamic view revealed that certain assets (like defense ETFs) became more synchronized post-invasion, while others (like USD/RUB and MOEX) showed increasing divergence.

5.5. Sector-Level Impact Assessment

Assets were grouped into predefined macro sectors:

Sector	Assets Included
Main Indices	`moex`, `dax`, `spy`
Commodities/Currencies	`brnt`, `usd_rub`, `usd_uah`
Aerospace/Defense	`ita`, `ppa`, `jets`, `rhmg`, `mtxgn`, `air`, `tcfp`, `baes`, `hiae`, `praf`, `bara`
European Equities	`ieur`, `sxepex`, `fez`
Energy Sector	`xle`, `bp`, `eni`, `eqnr`, `shell`

Table 3: Data to Extract

For each group:

- Average return and volatility were compared pre/post-invasion
- Sector-level bar plots were generated for visual analysis

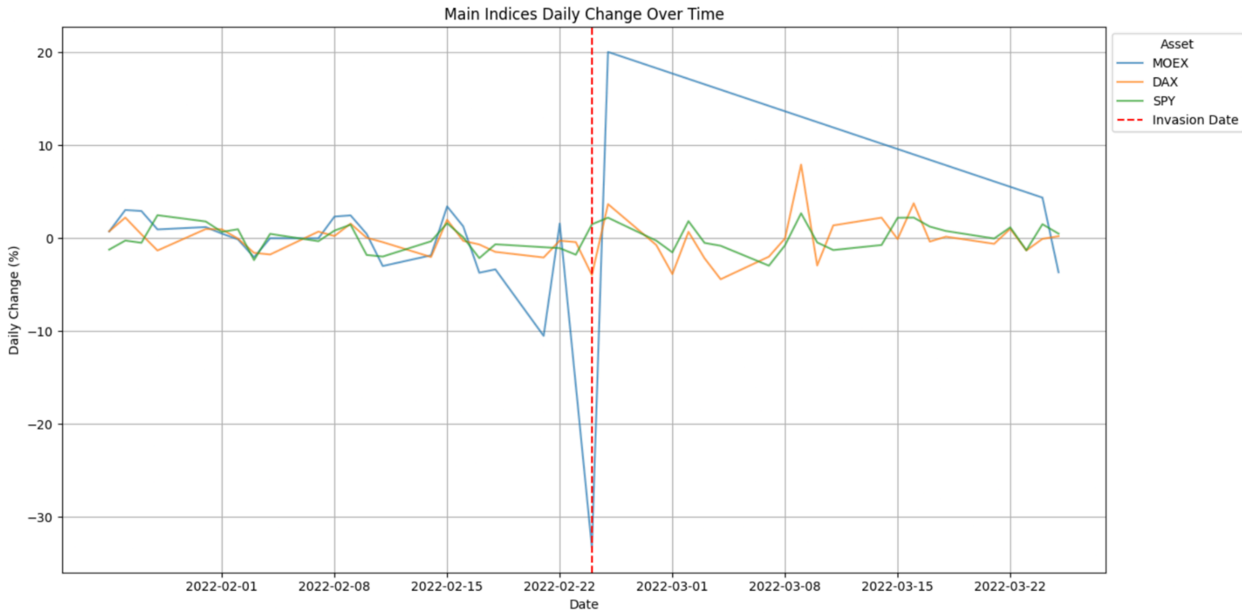


Figure 5.5 (a): Main Indices Daily Change Over Time

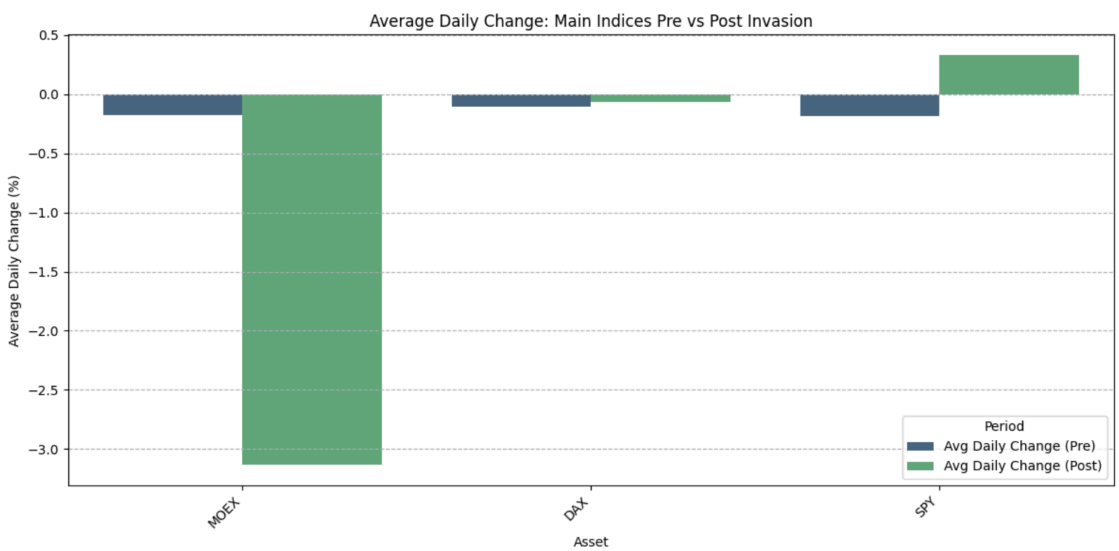


Figure 5.5 (b): Average Daily Change: Main Indices Pre vs Post Invasion

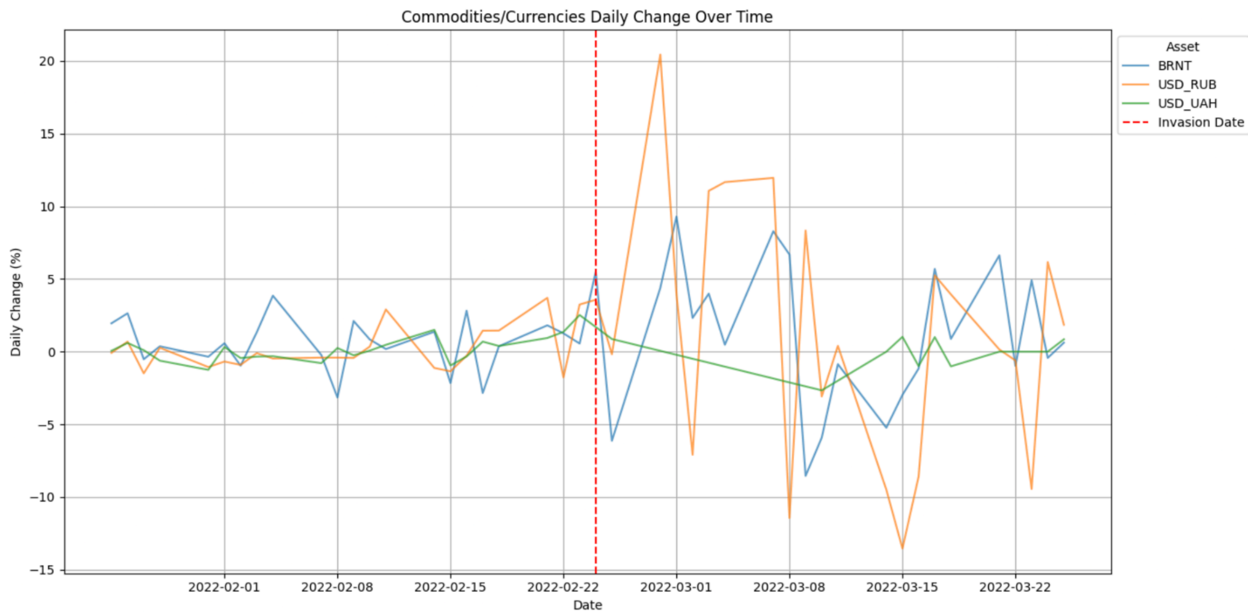


Figure 5.5 (c): Commodities/Currencies Daily Change Over Time

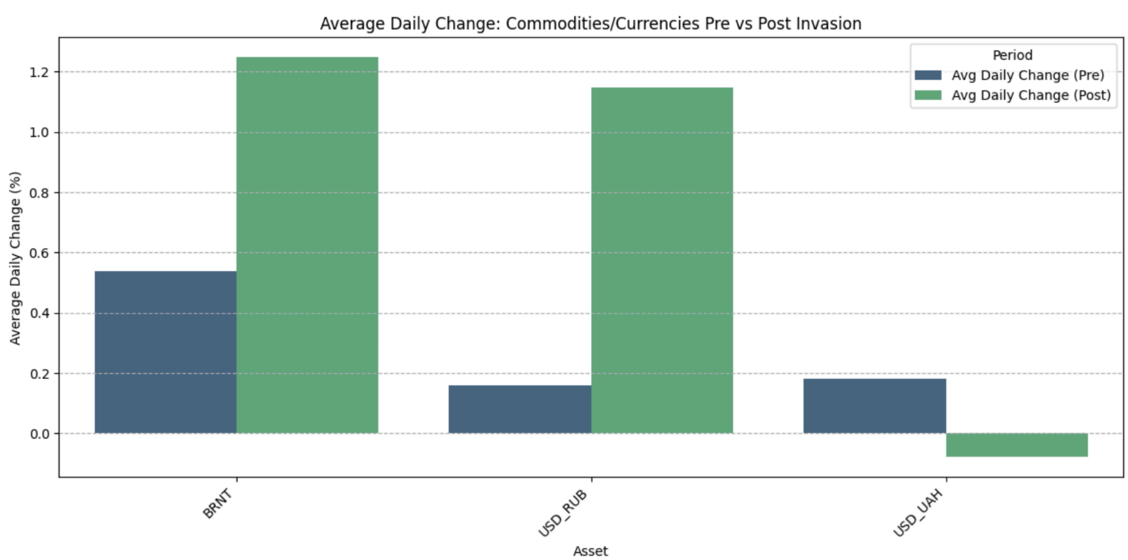


Figure 5.5 (d): Average Daily Change: Commodities/Currencies Pre vs Post Invasion

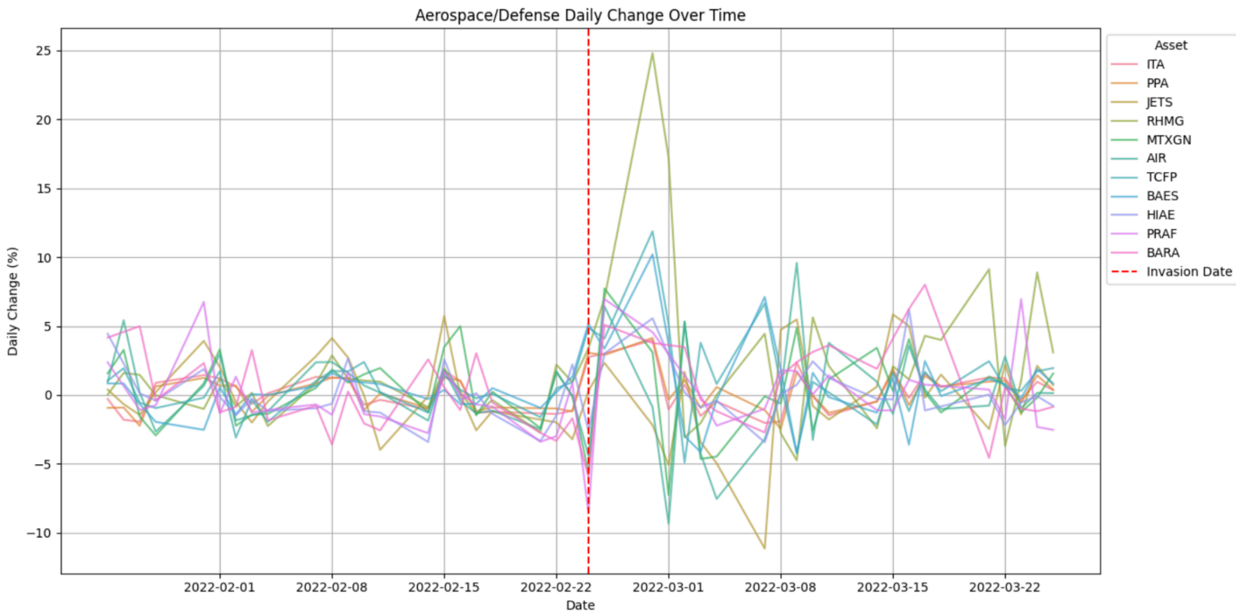


Figure 5.5 (e): Aerospace/Defense Daily Change Over Time

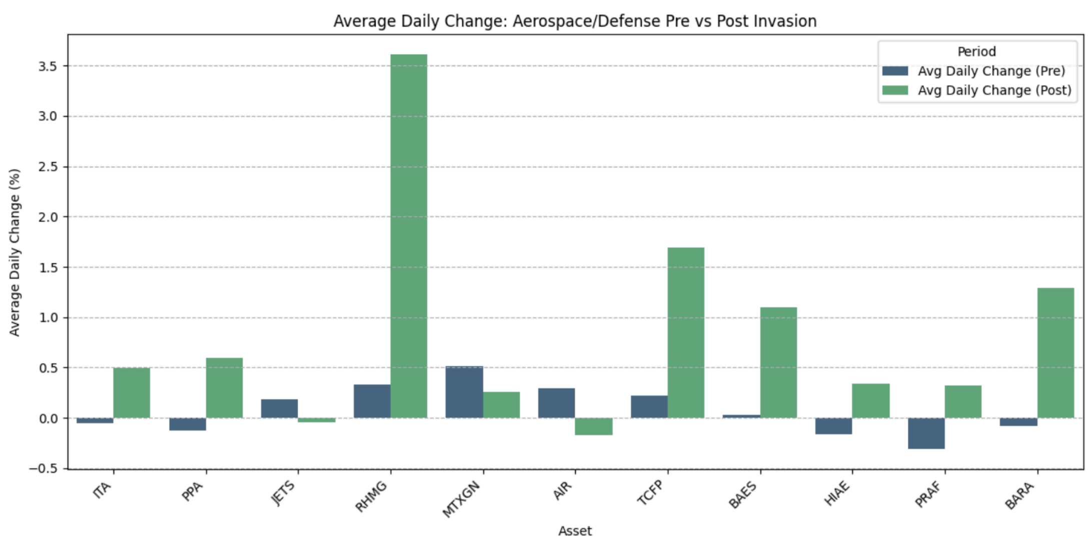


Figure 5.5 (f): Average Daily Change: Aerospace/Defense Pre vs Post Invasion

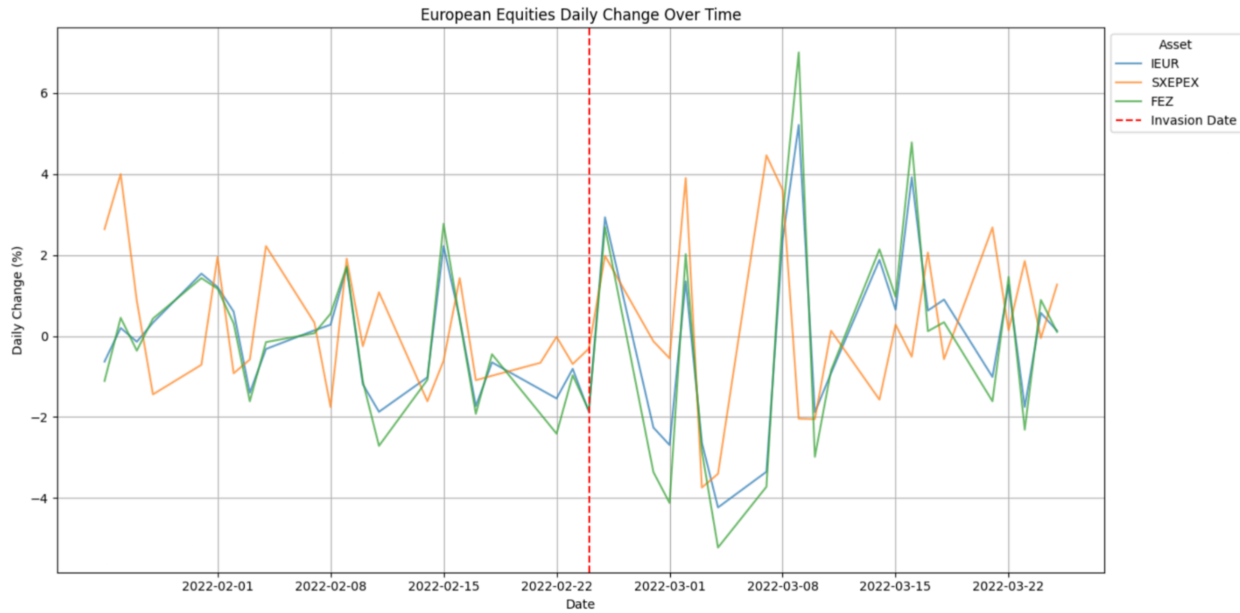


Figure 5.5 (g): European Equities Daily Change Over Time

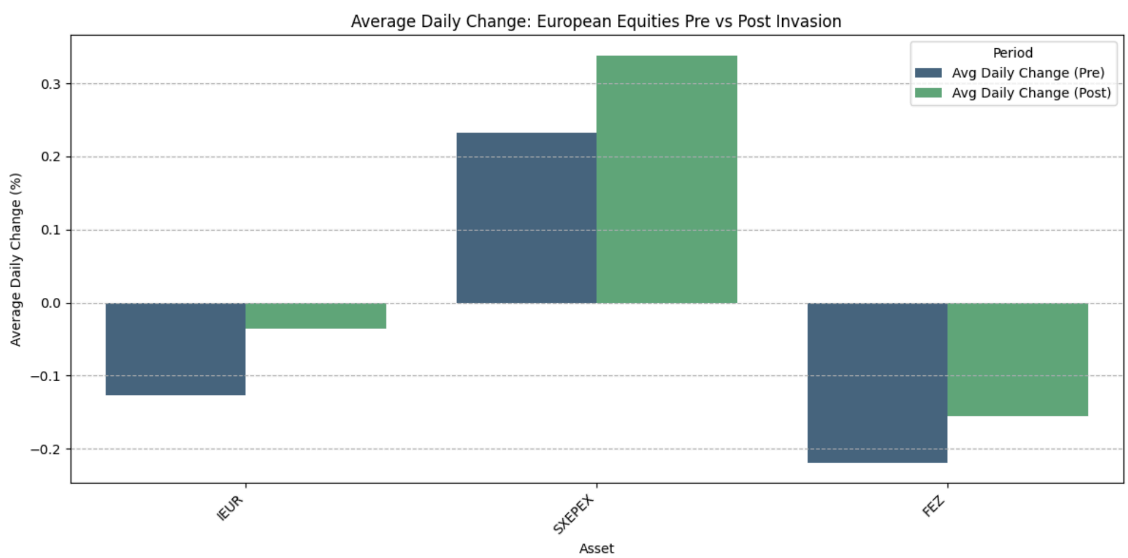


Figure 5.5 (h): Average Daily Change: European Equities Pre vs Post Invasion

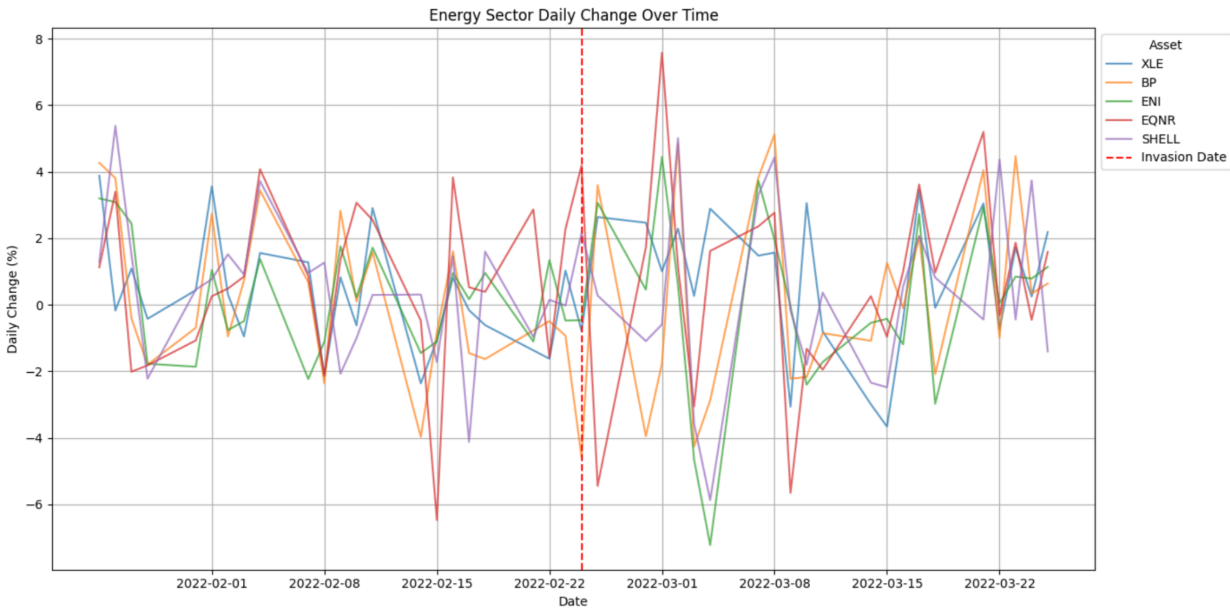


Figure 5.5 (i): Energy Sector Daily Change Over Time

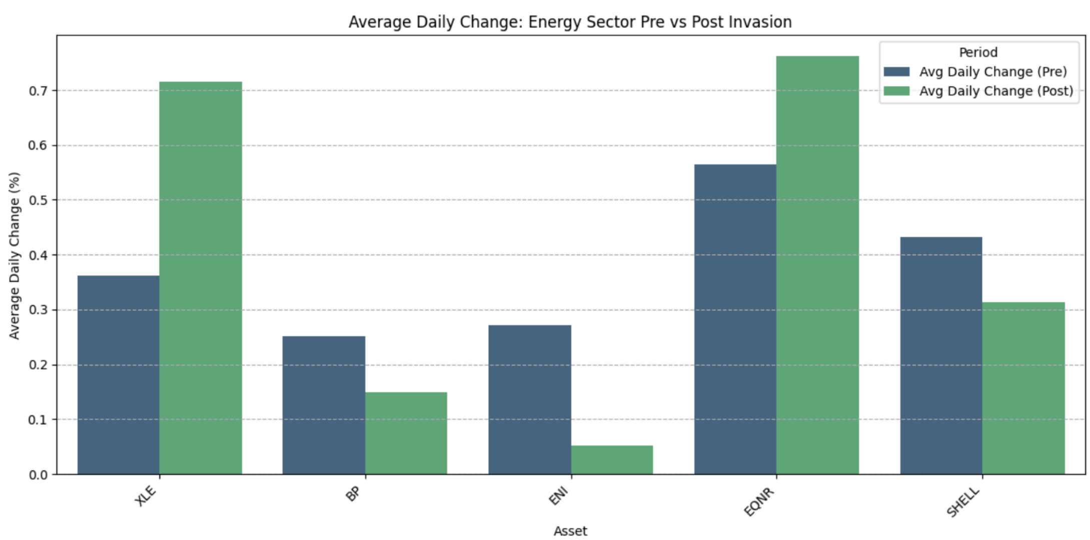


Figure 5.5 (j): Average Daily Change: Energy Sector Pre vs Post Invasion

This revealed that the Aerospace/Defense sector benefited most in terms of positive returns, while Energy and Currency sectors experienced high volatility.

5.6. Distribution Shifts Before vs. After Invasion

To examine how the behavioral distribution of returns and volatility changed:

- For each asset, histograms and KDE plots were created comparing:
    - Return distribution pre- vs. post-invasion
    - Volatility distribution pre- vs. post-invasion

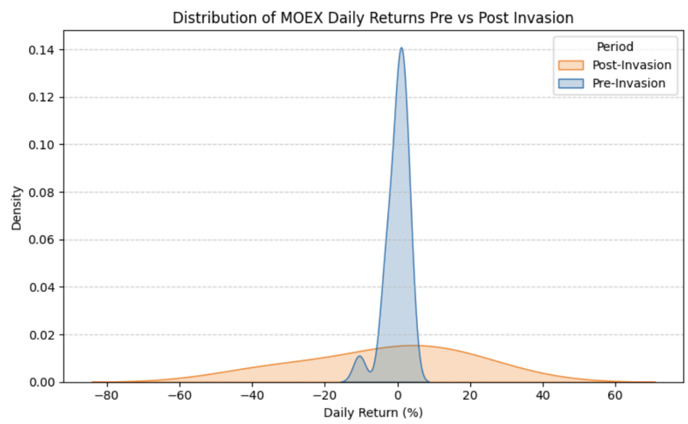


Figure 5.6 (a): Distribution of MOEX Daily Returns Pre vs Post Invasion

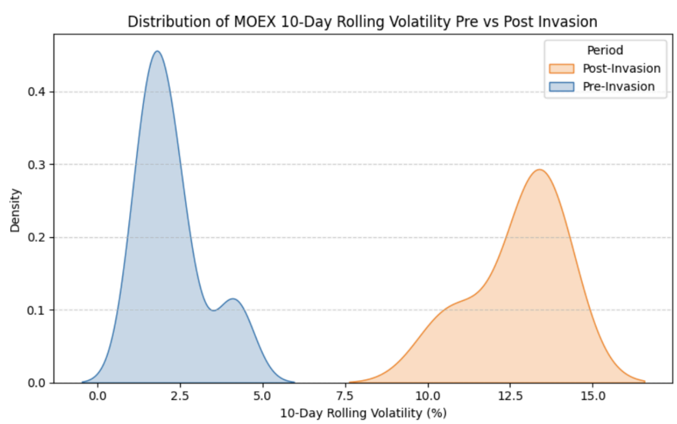


Figure 5.6 (b): Distribution of MOEX 10-Day Rolling Volatility Pre vs Post Invasion

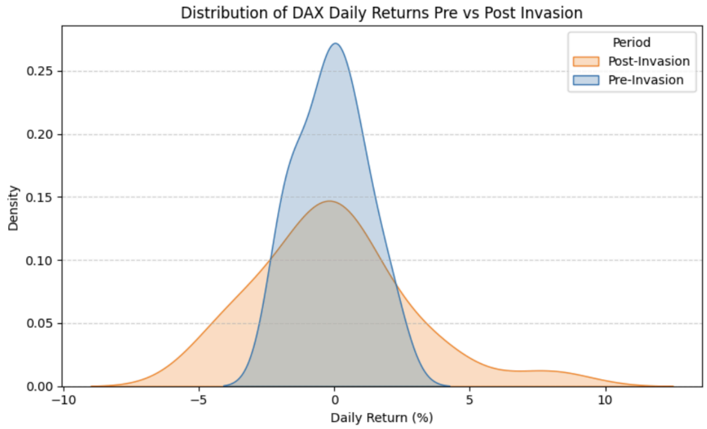


Figure 5.6 (c): Distribution of DAX Daily Returns Pre vs Post Invasion

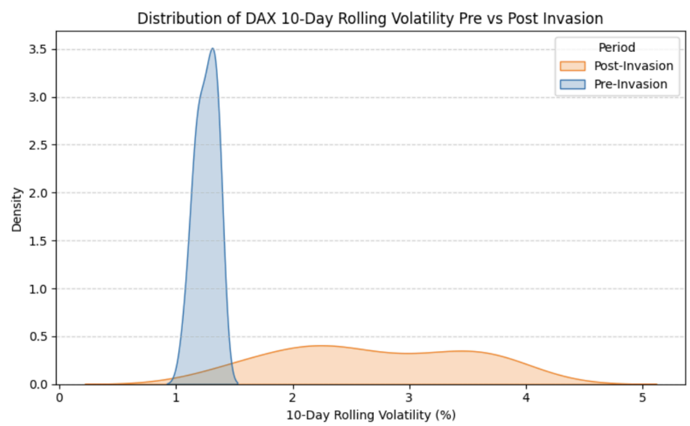


Figure 5.6 (d): Distribution of DAX 10-Day Rolling Volatility Pre vs Post Invasion

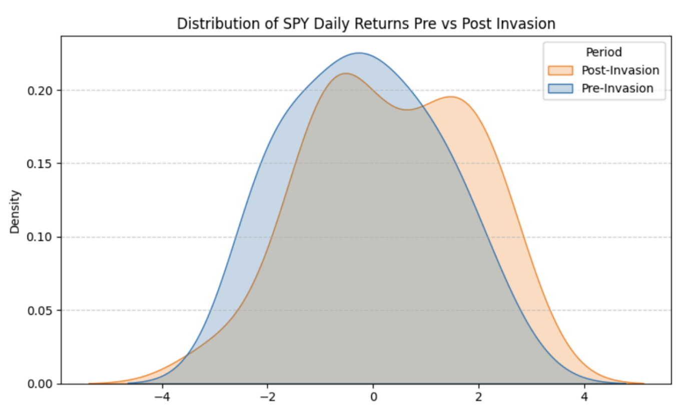


Figure 5.6 (e): Distribution of SPY Daily Returns Pre vs Post Invasion

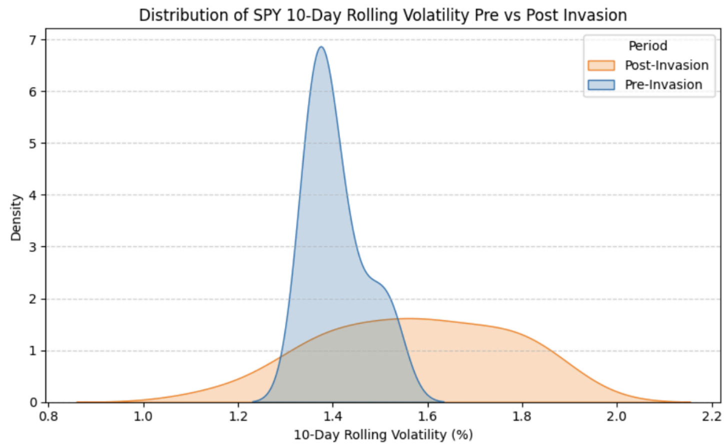


Figure 5.6 (f): Distribution of SPY 10-Day Rolling Volatility Pre vs Post Invasion

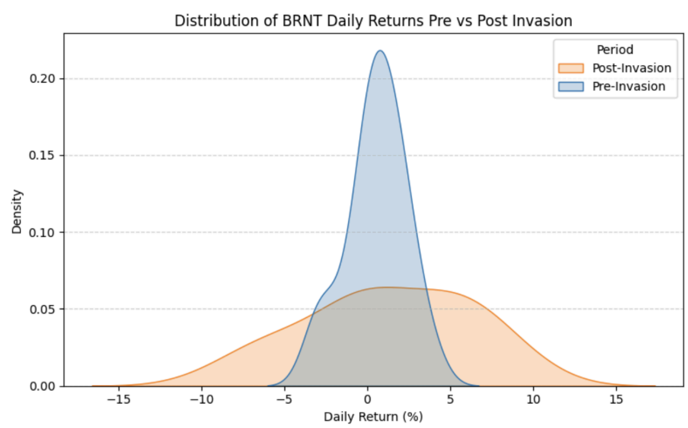


Figure 5.6 (g): Distribution of BRNT Daily Returns Pre vs Post Invasion

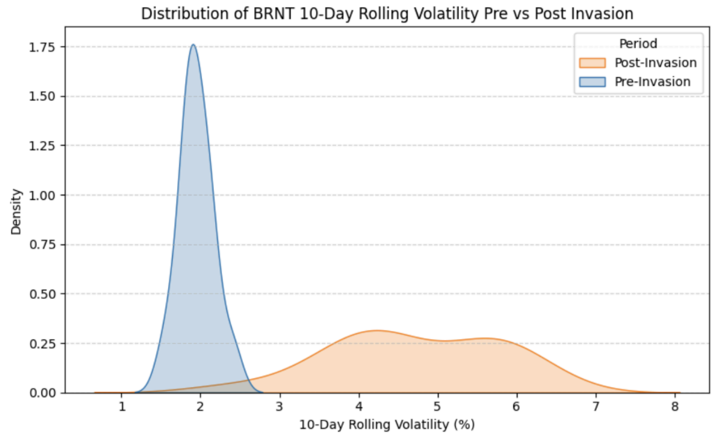


Figure 5.6 (h): Distribution of BRNT 10-Day Rolling Volatility Pre vs Post Invasion

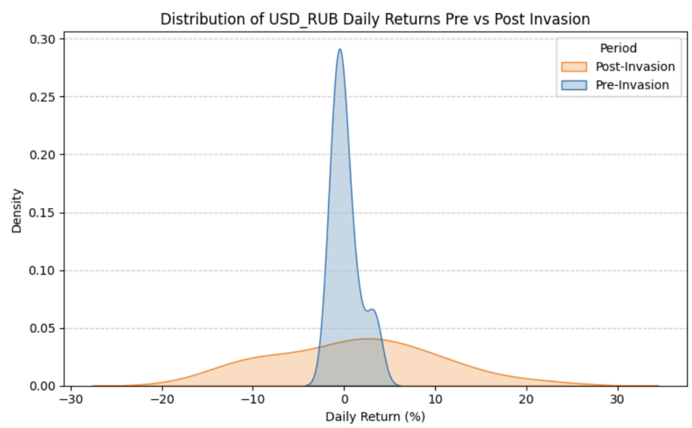


Figure 5.6 (i): Distribution of USD_RUB Daily Returns Pre vs Post Invasion

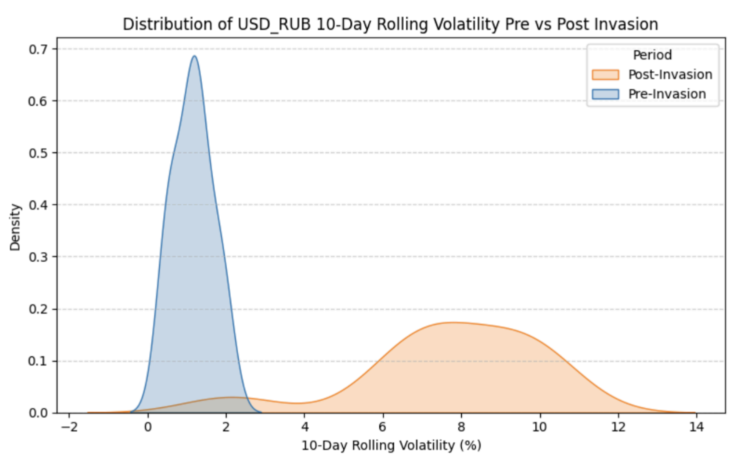


Figure 5.6 (j): Distribution of USD_RUB 10-Day Rolling Volatility Pre vs Post Invasion

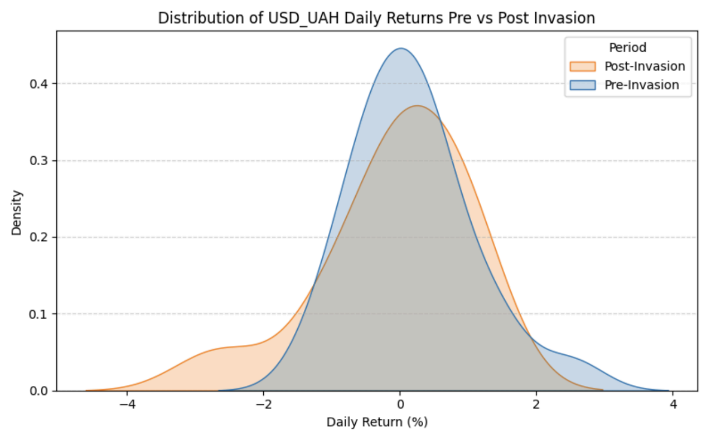


Figure 5.6 (k): Distribution of USD_UAH Daily Returns Pre vs Post Invasion

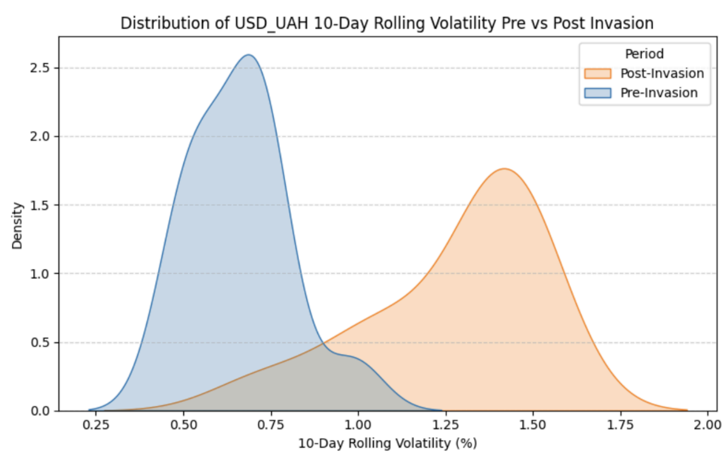


Figure 5.6 (l): Distribution of USD_UAH 10-Day Rolling Volatility Pre vs Post Invasion

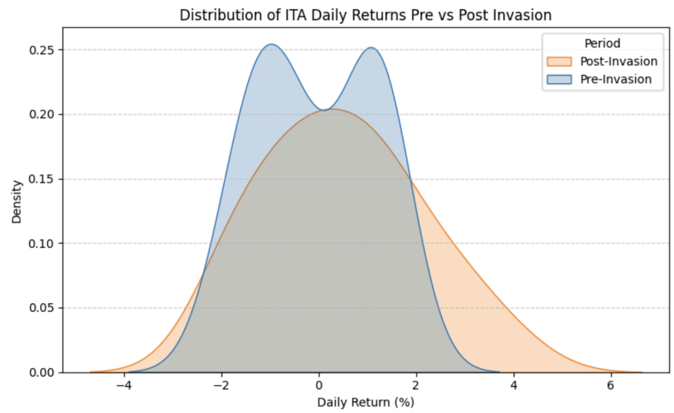


Figure 5.6 (m): Distribution of ITA Daily Returns Pre vs Post Invasion

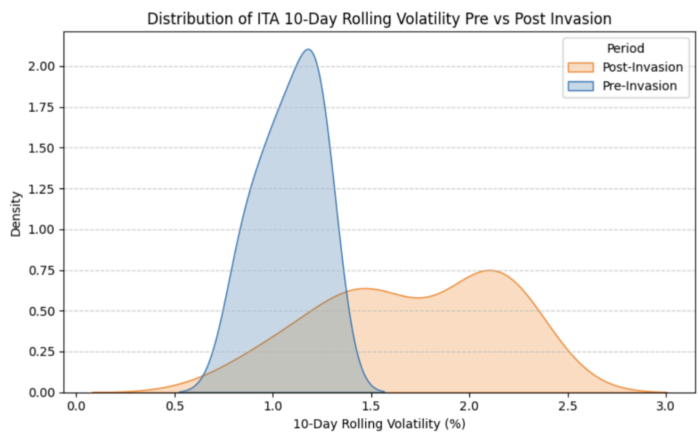


Figure 5.6 (n): Distribution of ITA 10-Day Rolling Volatility Pre vs Post Invasion

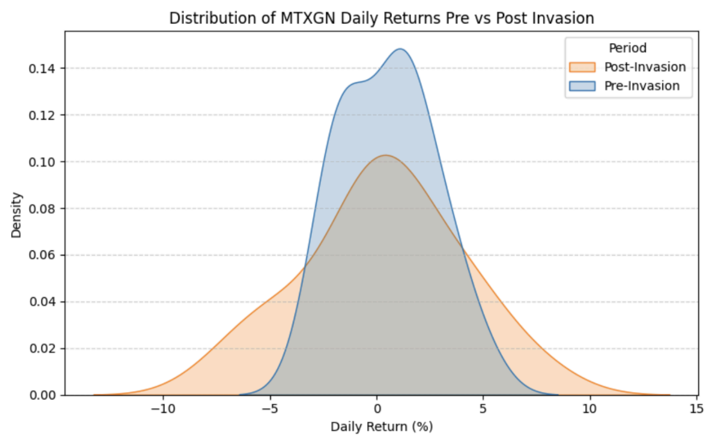


Figure 5.6 (o): Distribution of MTXGN Daily Returns Pre vs Post Invasion

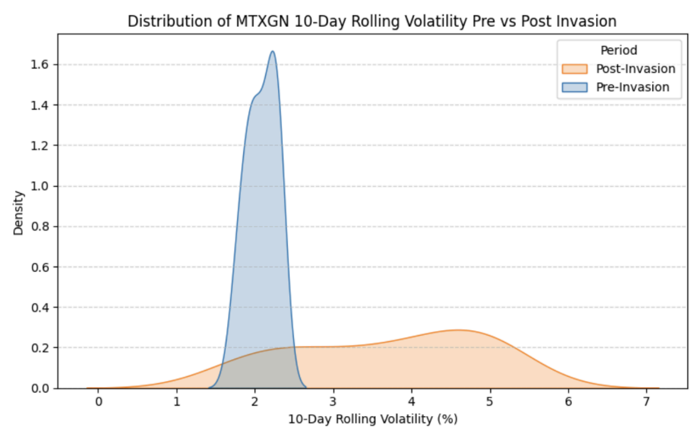


Figure 5.6 (p): Distribution of MTXGN 10-Day Rolling Volatility Pre vs Post Invasion

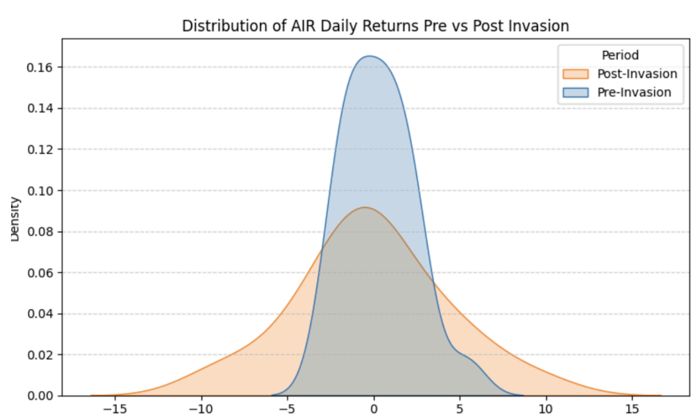


Figure 5.6 (q): Distribution of AIR Daily Returns Pre vs Post Invasion

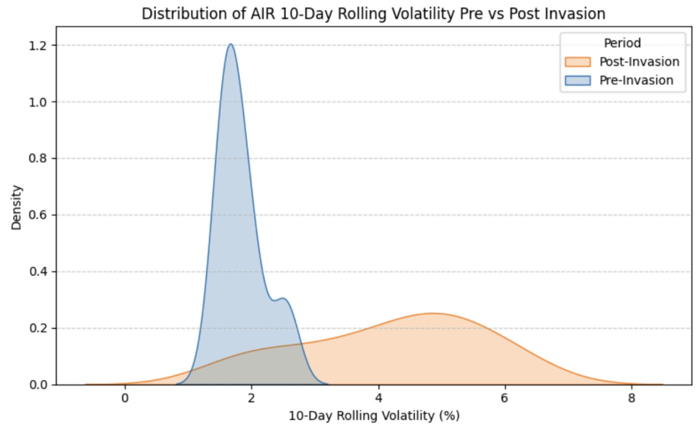


Figure 5.6 (r): Distribution of AIR 10-Day Rolling Volatility Pre vs Post Invasion

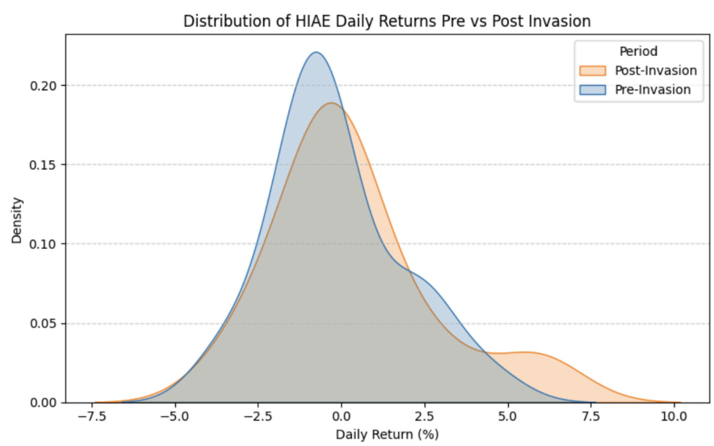


Figure 5.6 (s): Distribution of HIAE Daily Returns Pre vs Post Invasion

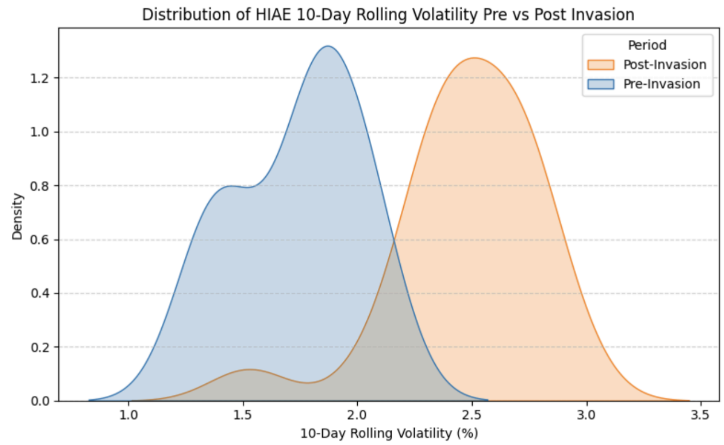


Figure 5.6 (t): Distribution of HIAE 10-Day Rolling Volatility Pre vs Post Invasion

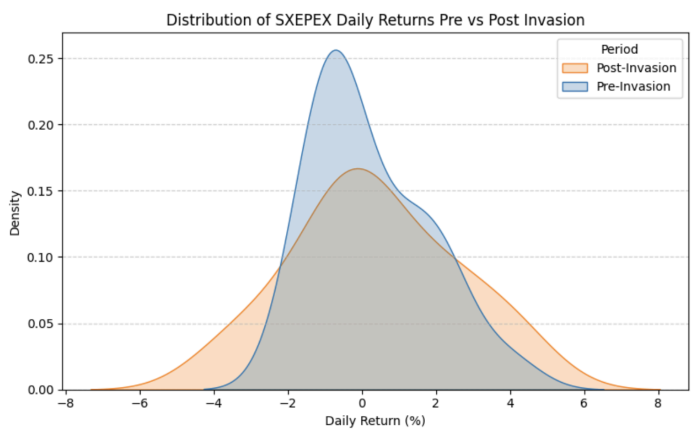


Figure 5.6 (u): Distribution of SXEPEX Daily Returns Pre vs Post Invasion

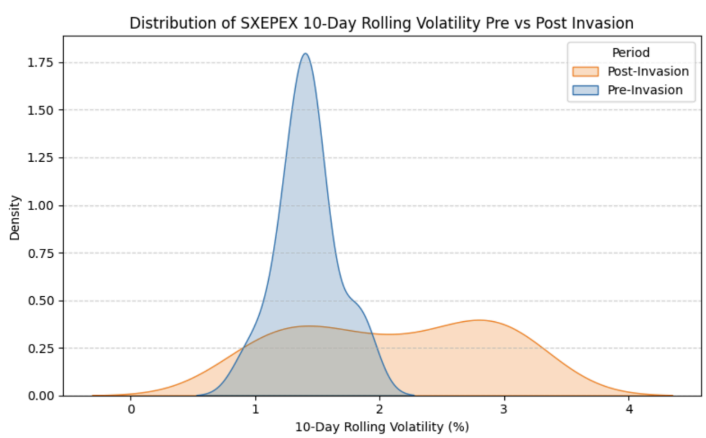


Figure 5.6 (v): Distribution of SXEPEX 10-Day Rolling Volatility Pre vs Post Invasion

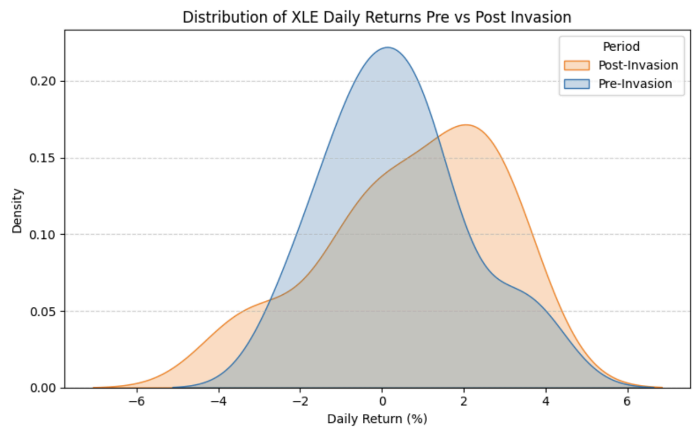


Figure 5.6 (w): Distribution of XLE Daily Returns Pre vs Post Invasion

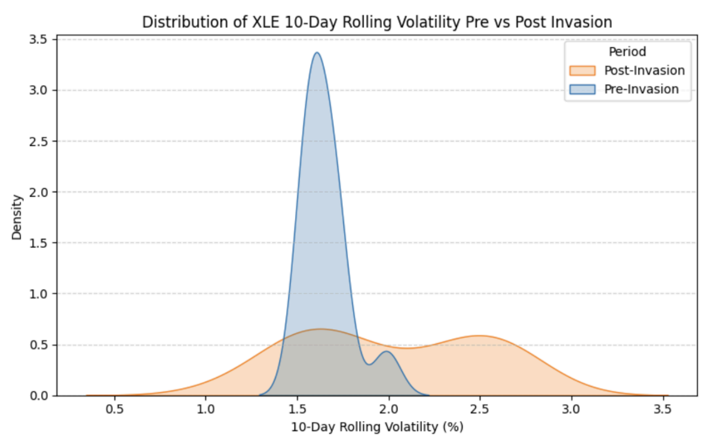


Figure 5.6 (x): Distribution of XLE 10-Day Rolling Volatility Pre vs Post Invasion

This reinforced that most indices and ETFs showed wider return dispersion and higher volatility post-invasion, consistent with panic and uncertainty in financial markets.

5.7. Dynamic Time Warping (DTW) & Lagged Correlation Analysis

To identify lagged or time-shifted relationships between assets:

- A reusable dtw_compare() function was created to compute DTW distance between any two Change_ series.
- A predefined list of economically meaningful pairs was analyzed (e.g., Brent → MOEX, USD/RUB → MOEX, ITA → SPY).
- For each pair:
    - DTW distance was computed
    - A visual plot of DTW alignment paths was generated

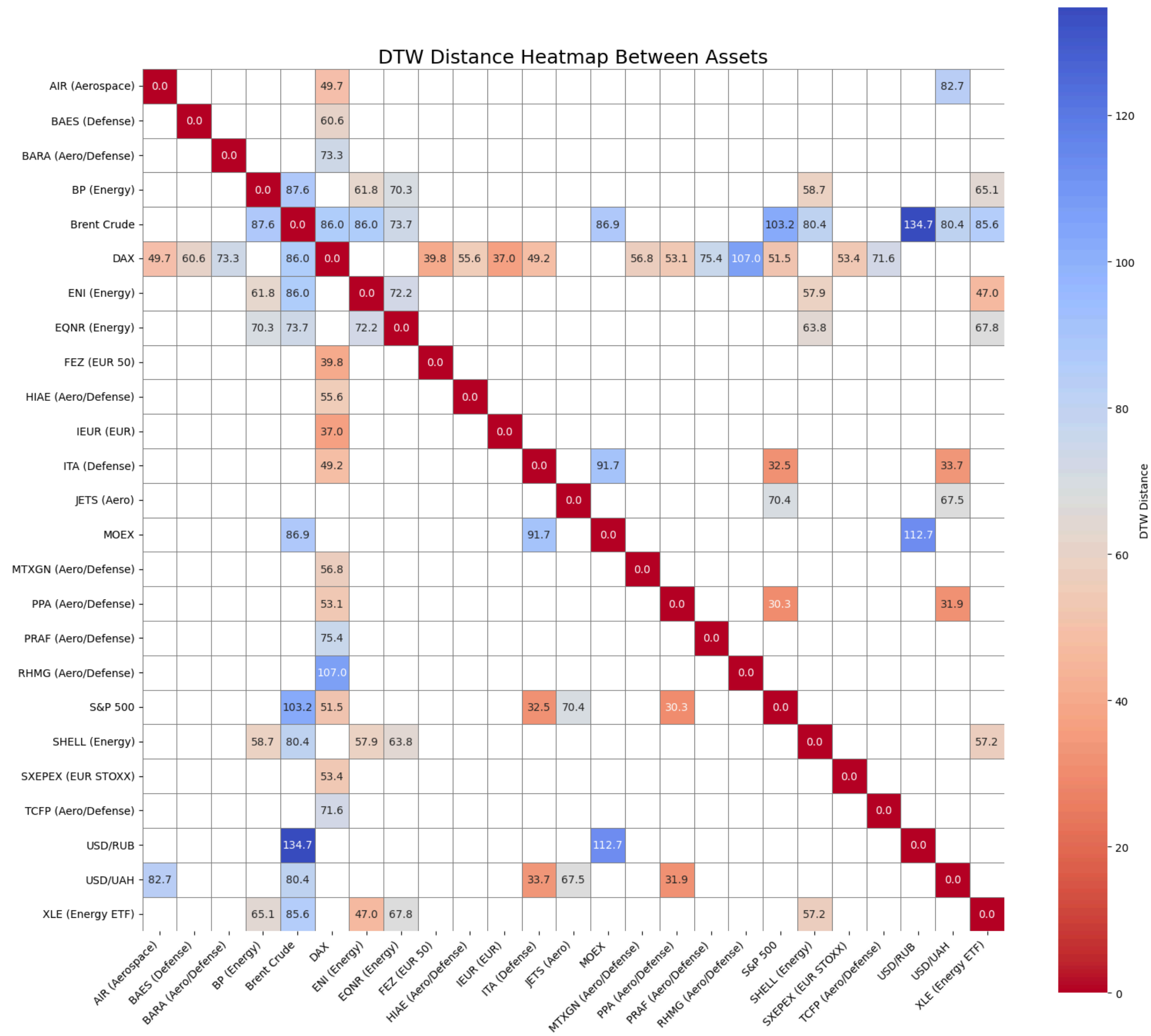


Figure 5.7 (a): DTW Distance Heatmap Between Assets

This uncovered strong lead-lag behavior, especially Brent leading MOEX, and ITA leading broader U.S. markets, reflecting reactive investor sentiment and delayed pricing-in of war-related risks.

5.8. DTW Distance Matrix and Clustering

To synthesize DTW results:

- All pairwise DTW distances were stored in a symmetric distance matrix
- A clustered heatmap was generated using hierarchical clustering
- Color-coded sector annotations were added to group assets visually

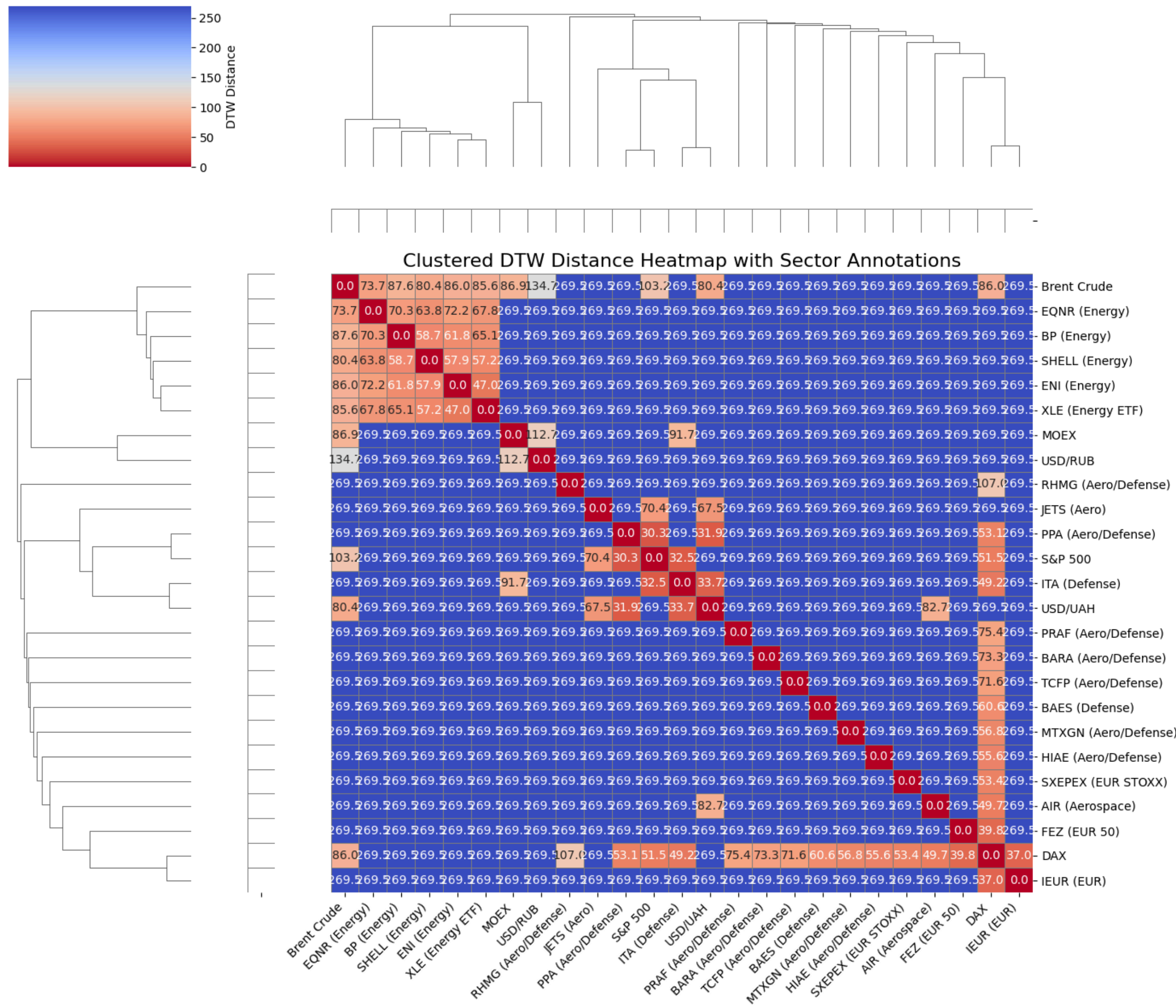


Figure 5.8 (a): Clustered DTW Distance Heatmap with Sector Annotations

This visual summary clearly showed:

- A tight cluster of defense ETFs and U.S. indices
- A separate block for currencies and commodities
- More disconnected behavior for emerging market equities and energy stocks

The clustering analysis helped map behavioral segmentation of markets during crises.

6. Conclusion

This project investigated the impact of the Russia–Ukraine conflict on global financial markets through a multi-faceted data analytics approach using historical price data, volatility analysis, and Dynamic Time Warping (DTW). The analysis spanned across major equity indices, defense and aerospace ETFs, energy and currency markets, and select regional sectoral representatives. The key takeaways include:

- Volatility Spikes and Structural Breaks: The onset of the conflict (Feb 24, 2022) coincided with sharp increases in volatility across equity indices, particularly in MOEX (Russia), Brent Crude, and defense sector ETFs, reflecting heightened uncertainty and geopolitical risk pricing.
- Pre- vs Post-Invasion Analysis revealed:
  - Russian assets (MOEX, RUB) experienced severe drawdowns and decoupled behavior post-invasion.
  - Defense and aerospace sectors (ITA, PPA, European defense stocks) demonstrated resilience and even gains, reflecting increased defense spending and geopolitical tailwinds.
  - Broader equity markets (S\&P 500, DAX) were impacted moderately and showed partial recovery post-invasion.
- DTW-Based Lead-Lag Analysis highlighted:
  - Strong dynamic alignment between U.S. defense ETFs and SPY, indicating synchronous movement.
  - Brent Crude's returns showed a leading behavior towards MOEX and European markets, although the alignment was imperfect due to structural breaks and sanctions.
  - USD/RUB and MOEX showed weak alignment, suggesting that currency and equity markets in Russia diverged in behavior post-conflict, likely due to capital controls and policy interventions.
- Clustering of DTW Distances showed natural groupings:

- U.S. defense, aerospace, and indices formed a tightly knit cluster.
- Commodities and currencies formed a separate cluster, more distantly related to equities.
- European and energy assets showed moderate connectivity, highlighting regional dependency on Russian resources.

7. Strategic Implications

- For Investors:
  - Sector rotation into defense and aerospace provided hedging opportunities during geopolitical escalation.
  - Commodities, especially oil and wheat, behaved as crisis hedges, but with volatility and timing complexity.
  - Traditional market correlations broke down, emphasizing the need for dynamic, event-driven portfolio rebalancing.
- For Policymakers:
  - The decoupling of MOEX and USD/RUB indicates that capital controls and sanctions have significant impacts on market behavior — sometimes detaching financial fundamentals from actual risk.
  - The alignment of defense stocks globally suggests a coordinated defense revaluation, providing insights into public spending sentiment and global military-industrial signals.
- For Risk Managers and Analysts:
  - Static correlation matrices are insufficient during crises — DTW and rolling volatility analysis provide better tools to model market dynamics under shock.
  - Geopolitical event modeling should integrate defense, energy, and currency interactions, especially across regional blocks (U.S., Europe, India).

Appendix (Screenshots of Bloomberg, Raw Data)

Screenshots from Bloomberg Terminal showing use of GP, HIST, CN, and GIP functions

Figure 1: GP MOEX

Figure 2: GP SPY

Figure 3: GP DAX

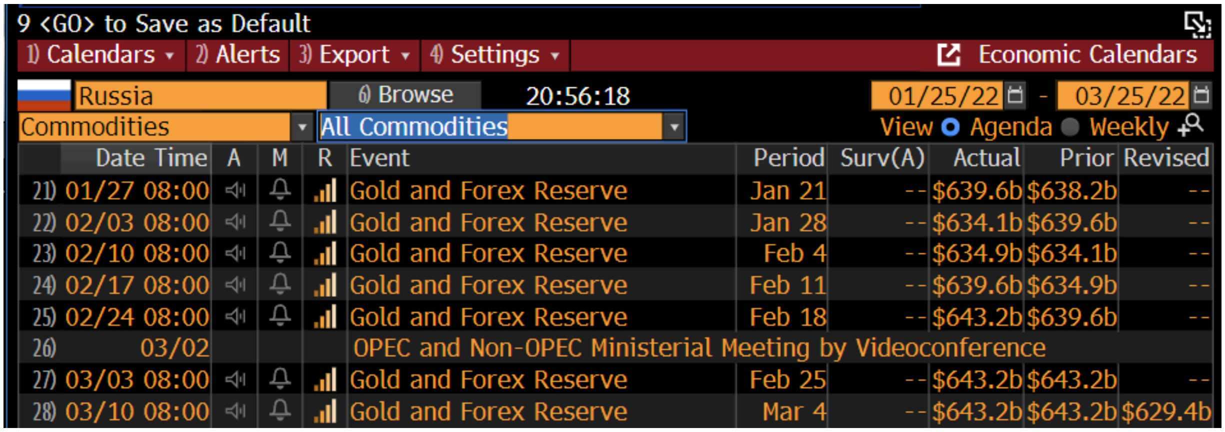
Figure 4: Commodity Crude Oil (Brent)

Figure 5: USD-RUB Currency Comparison

Figure 6: USD-UAH Currency Comparison

Figure 7: Brent Crude vs MOEX Comparison

Figure 8: ETCS Russia



9 <GO> to Save as Default
1) Calendars 2) Alerts 3) Export 4) Settings Economic Calendars
Russia 0) Browse 20:56:18 01/25/22 - 03/25/22
Commodities All Commodities View Agenda Weekly

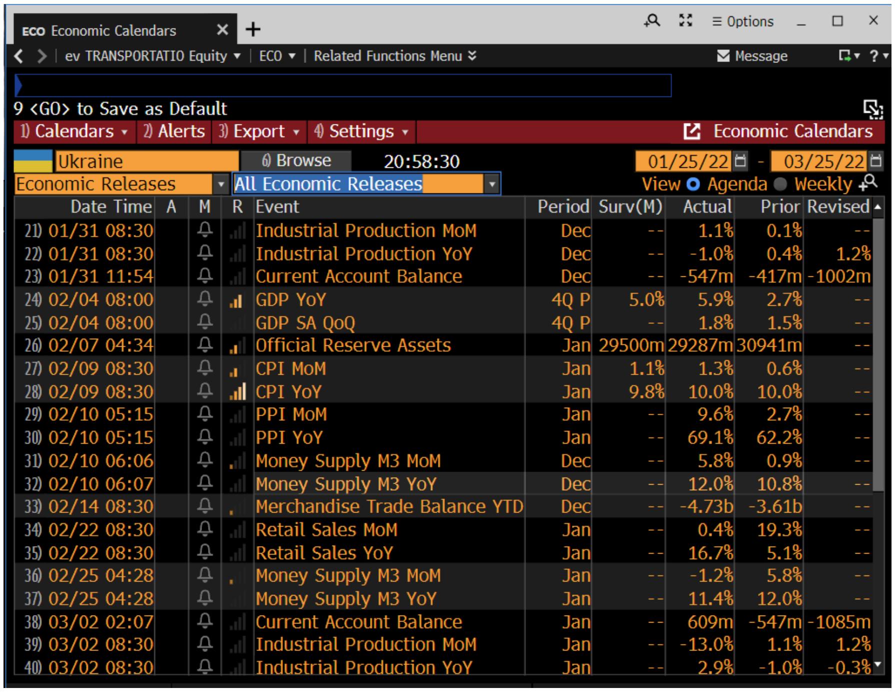
	Date Time	Event	Period	Surv(A)	Actual	Prior	Revised
21)	01/27 08:00	Gold and Forex Reserve	Jan 21	--	$639.6b	$638.2b	--
22)	02/03 08:00	Gold and Forex Reserve	Jan 28	--	$634.1b	$639.6b	--
23)	02/10 08:00	Gold and Forex Reserve	Feb 4	--	$634.9b	$634.1b	--
24)	02/17 08:00	Gold and Forex Reserve	Feb 11	--	$639.6b	$634.9b	--
25)	02/24 08:00	Gold and Forex Reserve	Feb 18	--	$643.2b	$639.6b	--
26)	03/02	OPEC and Non-OPEC Ministerial Meeting by Videoconference					
27)	03/03 08:00	Gold and Forex Reserve	Feb 25	--	$643.2b	$643.2b	--
28)	03/10 08:00	Gold and Forex Reserve	Mar 4	--	$643.2b	$643.2b	$629.4b

Figure 9: WECO (Commodities) Russia

ECO Economic Calendars Options
ev TRANSPORTATIO Equity ECO Related Functions Menu Message
9 <GO> to Save as Default
1) Calendars 2) Alerts 3) Export 4) Settings Economic Calendars
Russia 0) Browse 20:56:52 01/25/22 - 03/25/22
All Types All Types View Agenda Weekly

	Date Time	Event	Period	Surv(M)	Actual	Prior	Revised
21)	01/26 07:00	Russia Cancels OFZ Bonds Auction on 26 Jan.					
22)	01/26 11:00	PPI MoM	Dec	1.3%	0.8%	2.7%	--
23)	01/26 11:00	PPI YoY	Dec	29.0%	28.5%	29.2%	29.2%
24)	01/26 11:00	CPI WoW	Jan 21	--	--	0.10%	--
25)	01/26 11:00	CPI Weekly YTD	Jan 21	--	0.88%	0.66%	--
26)	01/27 08:00	Gold and Forex Reserve	Jan 21	--	$639.6b	$638.2b	--
27)	01/28 03:00	Money Supply Narrow Def	Jan 21	--	14.38t	14.39t	--
28)	01/28 03:30	SURVEY REPORT: Russia Economic Forecasts in January 2022					
29)	02/01 01:00	S&P Global Russia Manufacturi...	Jan	51.0	--	--	--
30)	02/02 07:00	Russia Sells 74.734B Rubles 2029 OFZ Bonds; Bid-Cover 1.38					
31)	02/02 11:00	Industrial Production YoY	Dec	4.5%	6.1%	7.0%	7.8%
32)	02/02 11:00	CPI WoW	Jan 28	--	0.19%	0.22%	--
33)	02/02 11:00	CPI Weekly YTD	Jan 28	--	1.07%	0.88%	--
34)	02/03 01:00	S&P Global Russia Services P...	Jan	49.5	--	--	--
35)	02/03 01:00	S&P Global Russia Composite ...	Jan	--	--	--	--
36)	02/03 08:00	Gold and Forex Reserve	Jan 28	--	$634.1b	$639.6b	--
37)	02/04 03:00	Money Supply Narrow Def	Jan 28	--	14.31t	14.38t	--
38)	02/04 04:16	Light Vehicle Car Sales YoY	Jan	-13.3%	-3.7%	-19.9%	--
39)	02/07 08:00	Official Reserve Assets	Jan	640.0b	630.2b	630.6b	--
40)	02/09 07:00	Russia Sells 20.657B Rubles of 7% 2036 OFZ Bonds; Bid-Cover...					

Figure 10: WECO (All Types) Russia



Date Time	A	M	R	Event	Period	Surv(M)	Actual	Prior	Revised
21) 01/31 08:30				Industrial Production MoM	Dec	--	1.1%	0.1%	--
22) 01/31 08:30				Industrial Production YoY	Dec	--	-1.0%	0.4%	1.2%
23) 01/31 11:54				Current Account Balance	Dec	--	-547m	-417m	-1002m
24) 02/04 08:00				GDP YoY	4Q P	5.0%	5.9%	2.7%	--
25) 02/04 08:00				GDP SA QoQ	4Q P	--	1.8%	1.5%	--
26) 02/07 04:34				Official Reserve Assets	Jan	29500m	29287m	30941m	--
27) 02/09 08:30				CPI MoM	Jan	1.1%	1.3%	0.6%	--
28) 02/09 08:30				CPI YoY	Jan	9.8%	10.0%	10.0%	--
29) 02/10 05:15				PPI MoM	Jan	--	9.6%	2.7%	--
30) 02/10 05:15				PPI YoY	Jan	--	69.1%	62.2%	--
31) 02/10 06:06				Money Supply M3 MoM	Dec	--	5.8%	0.9%	--
32) 02/10 06:07				Money Supply M3 YoY	Dec	--	12.0%	10.8%	--
33) 02/14 08:30				Merchandise Trade Balance YTD	Dec	--	-4.73b	-3.61b	--
34) 02/22 08:30				Retail Sales MoM	Jan	--	0.4%	19.3%	--
35) 02/22 08:30				Retail Sales YoY	Jan	--	16.7%	5.1%	--
36) 02/25 04:28				Money Supply M3 MoM	Jan	--	-1.2%	5.8%	--
37) 02/25 04:28				Money Supply M3 YoY	Jan	--	11.4%	12.0%	--
38) 03/02 02:07				Current Account Balance	Jan	--	609m	-547m	-1085m
39) 03/02 08:30				Industrial Production MoM	Jan	--	-13.0%	1.1%	1.2%
40) 03/02 08:30				Industrial Production YoY	Jan	--	2.9%	-1.0%	-0.3%

Figure 11: WECO (Economic Releases) Ukraine

Headline	Tickers	Source	Date
China Woos India as Both Face Western Ire Over Ukraine	80070MF	WPT	03/2022
North Korea has 'likely more in store' after missile test: US		AFP	03/2022
Malta PM heads for re-election despite corruption fears		AFP	03/2022
Russia signals change in war plans as 'first stage' siege in Kyiv slows		BDR	03/2022
Ukrainian forces launch counter-offensive in Russian-controlled Kherson	SBER RM	TEL	03/2022
Ukraine 'disappointed' in NATO, as Biden Visits U.S. Troops in Poland		WPT	03/2022
Satellite photos show Russian landing ship destroyed by Ukrainian forces as it attemp...		BDR	03/2022
Review -- Barron's	3333 HK...	DJ	03/2022
Review	3333 HK...	BRR	03/2022
Review & Preview: The Shadow Of the 1940s -- Barron's	BA,BRK/...	DJ	03/2022
Review & Preview: The Shadow Of the 1940s	BA,BRK/...	BRR	03/2022
U.S.-China Decoupling Is Under Way -- Barron's		DJ	03/2022
U.S.-China Decoupling Is Under Way		BRR	03/2022
Russia-Ukraine war: what we know on day 31 of the invasion		GUA	03/2022
Market View -- Barron's	BLK,WFC	DJ	03/2022
Market View	BLK,WFC	BRR	03/2022
Commodities: Oil Producers' Plan Will Keep Prices High -- Barron's		DJ	03/2022
Commodities: Oil Producers' Plan Will Keep Prices High		BRR	03/2022

Figure 12: CN (News Articles - 'Russia' & 'Ukraine')

Exported CSV samples of raw_files can be found at:

https://github.com/suletanmay/Market-Tremors/tree/master/raw_data

Project source code, cleaned dataset samples, and model notebooks can be found at:
https://github.com/suletanmay/Market-Tremors/tree/master

References

[1] Bloomberg Terminal Documentation Reuters

[2] CNBC, Bloomberg News (for geopolitical timeline references)

[3] Investing.com (for historical data)

[4] Tickertape.in (for Indian defence stocks)